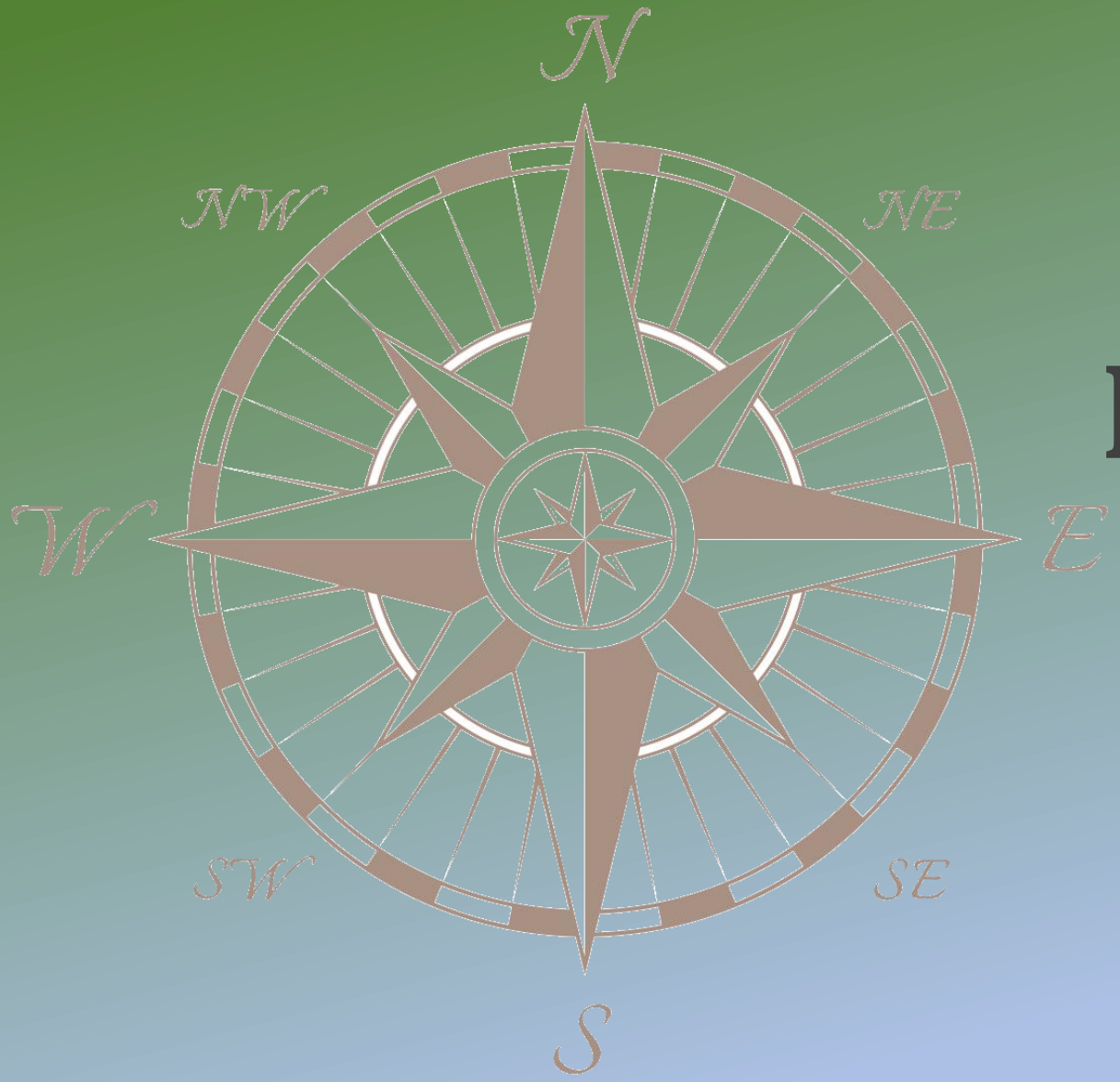


Learning and Understanding Metes and Bounds Descriptions

Sneak-peek into a Foreign Language

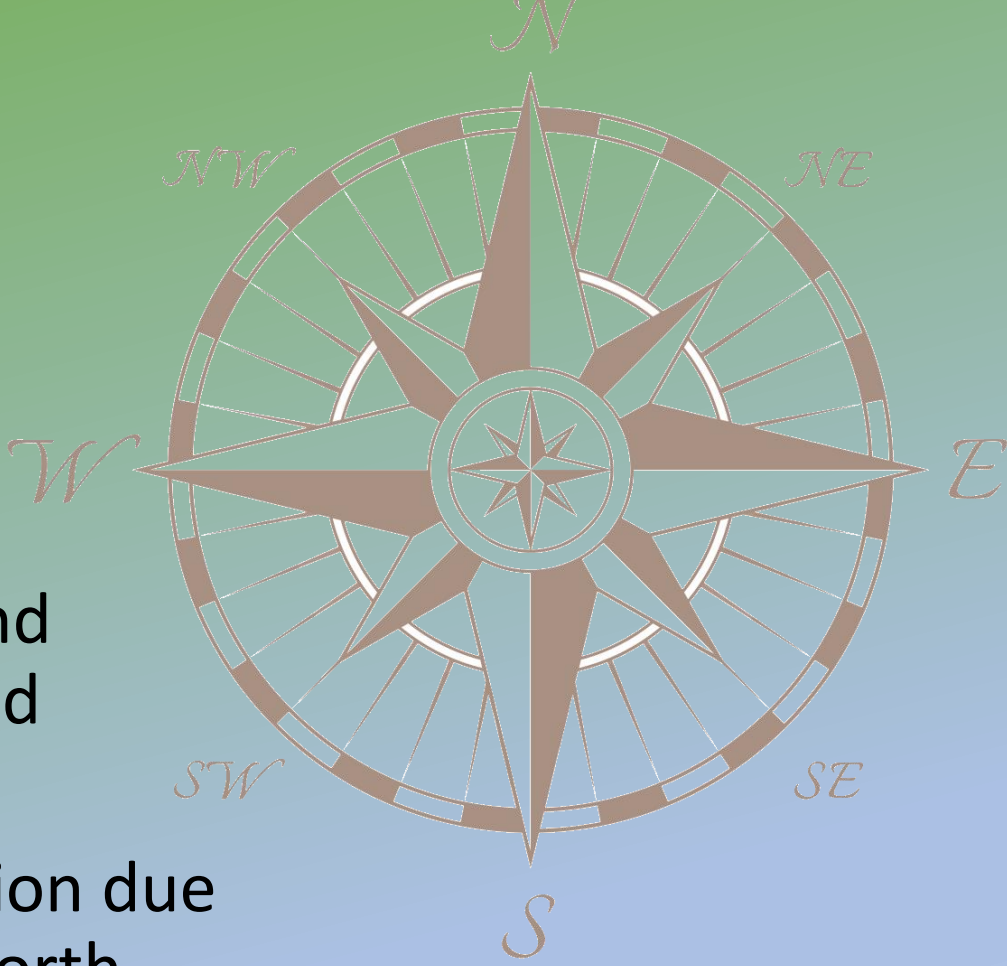
The Need:

Understand what land is going under conservation easement and what you are monitoring every year.



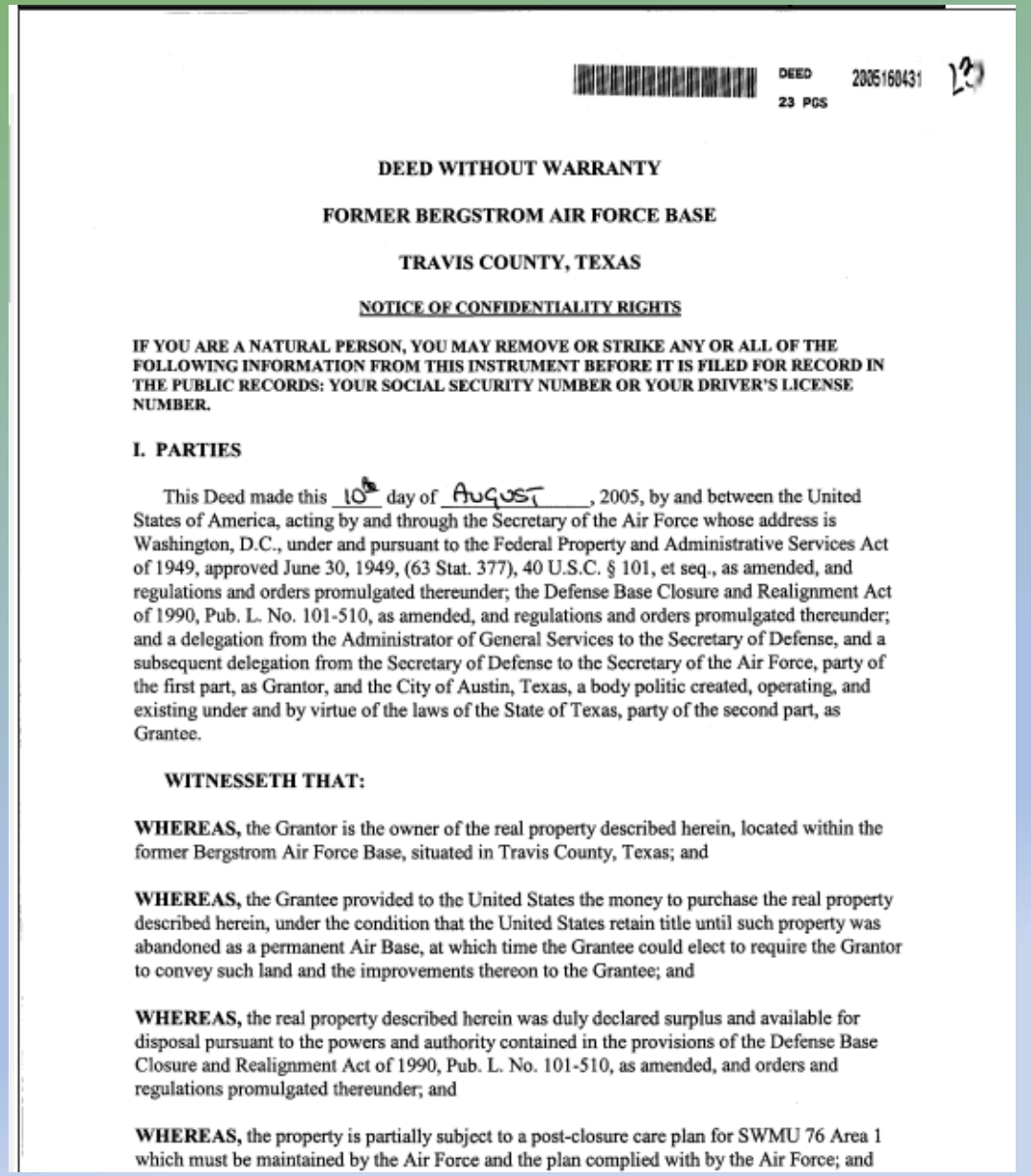
Objectives:

- basic understanding of distances, bearings, adjoiners and monuments
- ability to research necessary documents and produce an accurate map using a metes and bounds description
- recognize differences in boundary orientation due to coordinate systems and association to north



Scenarios - Real life examples:

- Recording Fail: M&B were recorded with a conservation easement placing the wrong half of the property under easement.
- The mistake was found, and a correction conservation easement was filed.
- If left unnoticed, the land trust would have continued monitoring the wrong half.



DEED 2005160431
23 PGS

DEED WITHOUT WARRANTY

FORMER BERGSTROM AIR FORCE BASE

TRAVIS COUNTY, TEXAS

NOTICE OF CONFIDENTIALITY RIGHTS

IF YOU ARE A NATURAL PERSON, YOU MAY REMOVE OR STRIKE ANY OR ALL OF THE FOLLOWING INFORMATION FROM THIS INSTRUMENT BEFORE IT IS FILED FOR RECORD IN THE PUBLIC RECORDS: YOUR SOCIAL SECURITY NUMBER OR YOUR DRIVER'S LICENSE NUMBER.

I. PARTIES

This Deed made this 10th day of August, 2005, by and between the United States of America, acting by and through the Secretary of the Air Force whose address is Washington, D.C., under and pursuant to the Federal Property and Administrative Services Act of 1949, approved June 30, 1949, (63 Stat. 377), 40 U.S.C. § 101, et seq., as amended, and regulations and orders promulgated thereunder; the Defense Base Closure and Realignment Act of 1990, Pub. L. No. 101-510, as amended, and regulations and orders promulgated thereunder; and a delegation from the Administrator of General Services to the Secretary of Defense, and a subsequent delegation from the Secretary of Defense to the Secretary of the Air Force, party of the first part, as Grantor, and the City of Austin, Texas, a body politic created, operating, and existing under and by virtue of the laws of the State of Texas, party of the second part, as Grantee.

WITNESSETH THAT:

WHEREAS, the Grantor is the owner of the real property described herein, located within the former Bergstrom Air Force Base, situated in Travis County, Texas; and

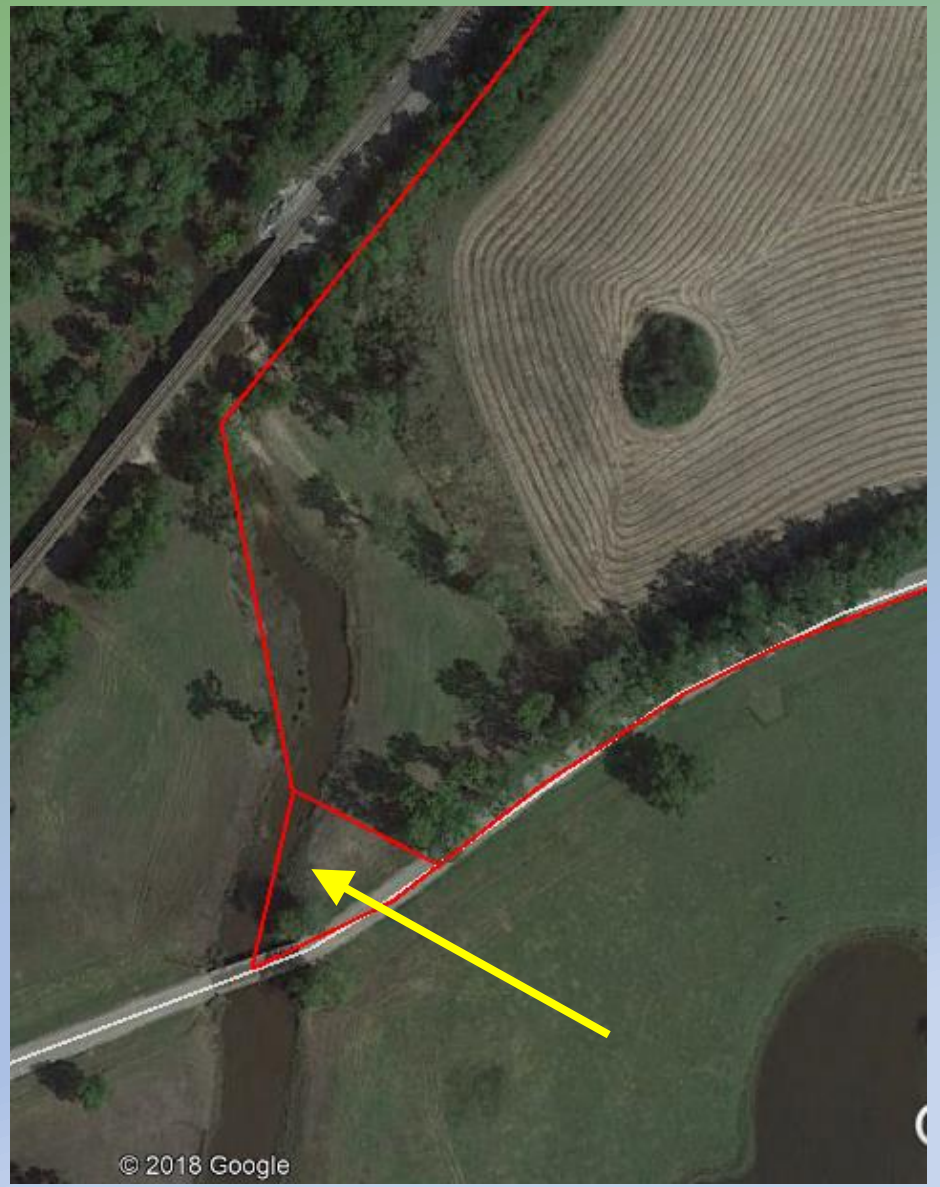
WHEREAS, the Grantee provided to the United States the money to purchase the real property described herein, under the condition that the United States retain title until such property was abandoned as a permanent Air Base, at which time the Grantee could elect to require the Grantor to convey such land and the improvements thereon to the Grantee; and

WHEREAS, the real property described herein was duly declared surplus and available for disposal pursuant to the powers and authority contained in the provisions of the Defense Base Closure and Realignment Act of 1990, Pub. L. No. 101-510, as amended, and orders and regulations promulgated thereunder; and

WHEREAS, the property is partially subject to a post-closure care plan for SWMU 76 Area 1 which must be maintained by the Air Force and the plan complied with by the Air Force; and

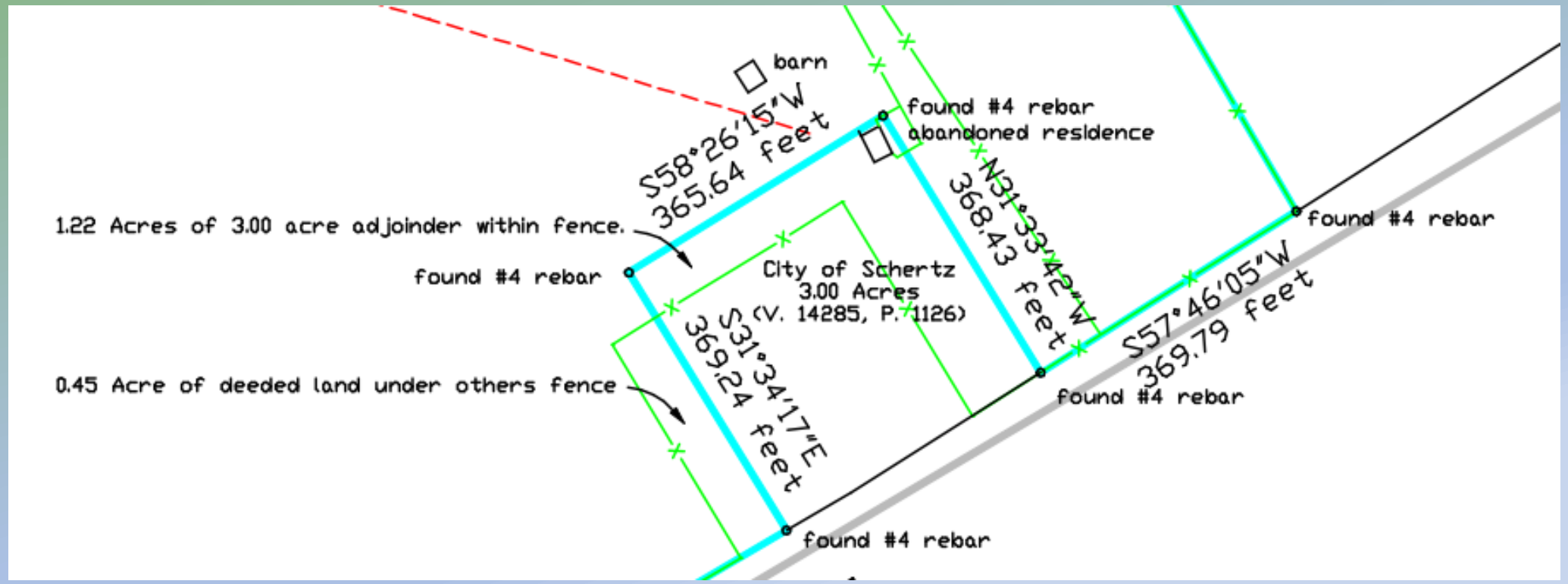
Scenarios - Real life examples:

- Mapping Fail: A map was produced showing a portion of the landowner's property as not being part of the CE.
- The landowner wanted to sell this portion of her property
- When the M&B were used to re-produce the map, that portion was under conservation easement and was subject to the restrictions of the easement.



Scenarios - Real life examples:

- Fencing Fail: A new property was proposed to go under CE. The survey showed the land owned by the grantor differed from what was within the fence.
- The Phase 1 was complete on the fenced area and did not evaluate the fill material deposited within the fence.
- The M&B were needed to know where the boundary lines actually were, not where the fence was.



Metes and Bounds:

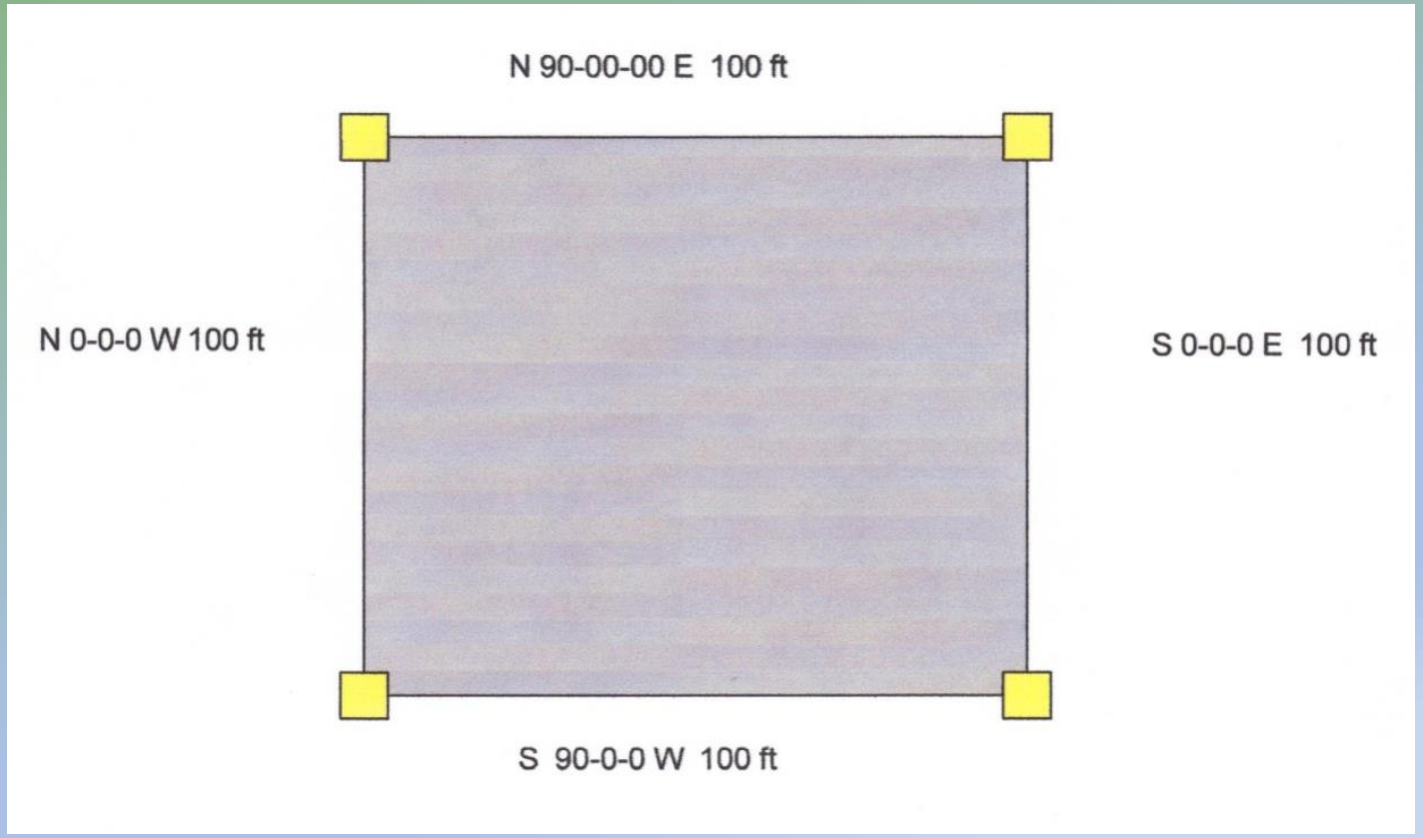
- Metes and Bounds Description

AKA

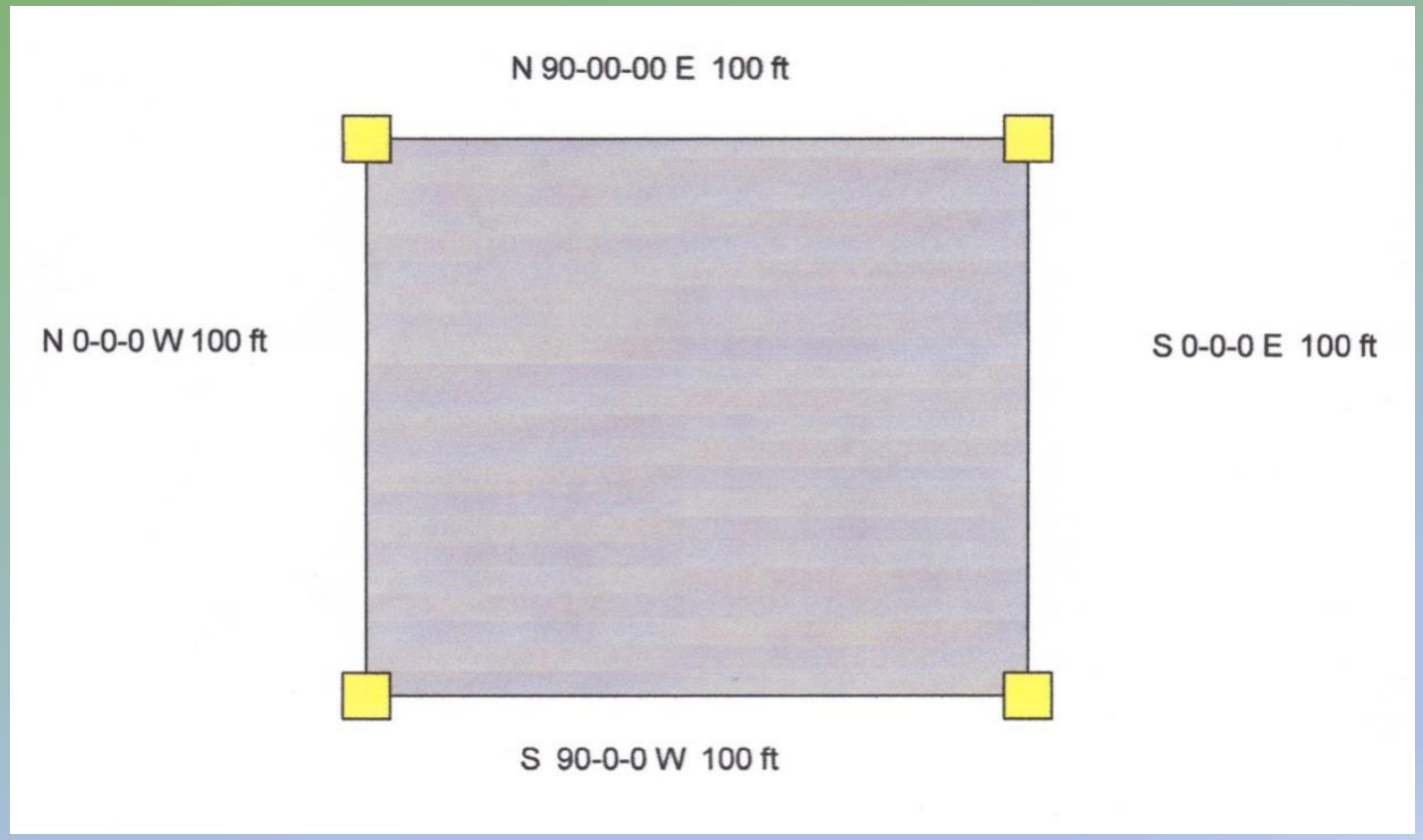
- Legal Description

AKA

- Field Notes

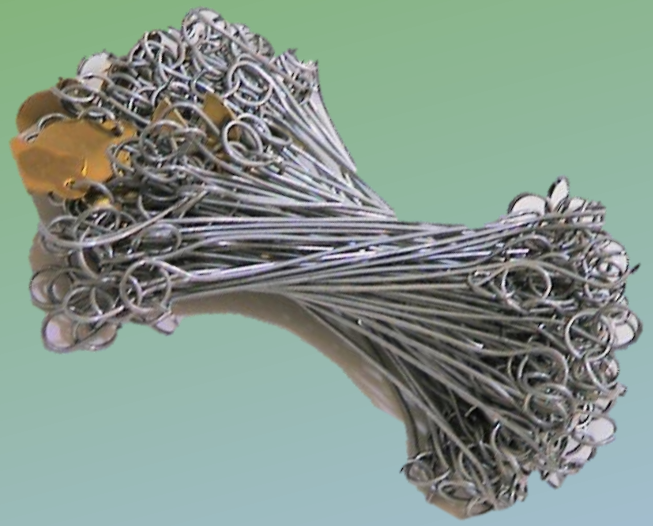


What are Metes:
Latin for measurements



Description of the bearings and distances of the perimeter of a property

What are Metes:
Types of Measurement



Standard (or "international") **foot**: .3048 meters

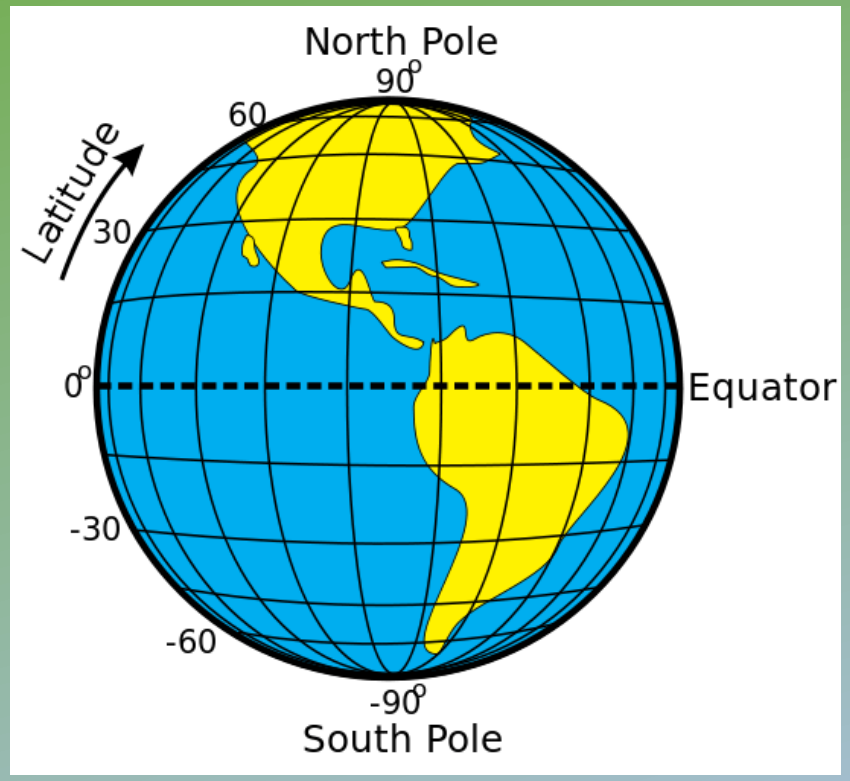
U.S. survey foot: 1200/3937 meters

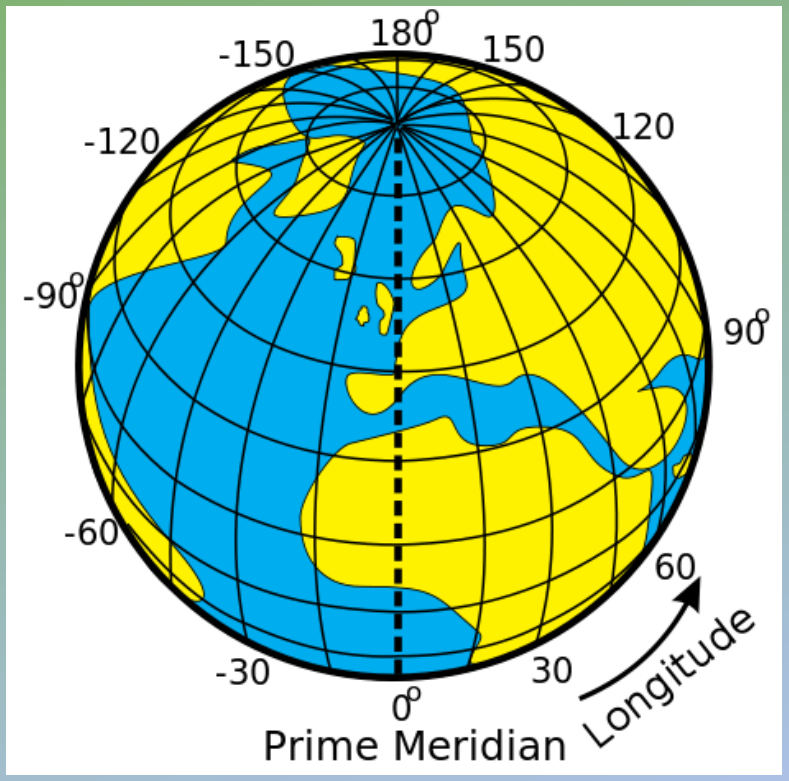
One is defined in relation to the meter by a decimal expression, the other by a fraction.
(2parts per million)

Texas Vara $33^{1}/_{3}$ inches 3 varas are exactly 100 inches,
GLO records

Gunter's Chain Edmund Gunter (1581–1626), 66 foot long measuring chain of 100 links.

What are Metes: Coordinates





<u>Degree minute second:</u>

Latitude: 30°12’30"N

Longitude: 97°39'57"W

<u>Decimal Degree:</u>

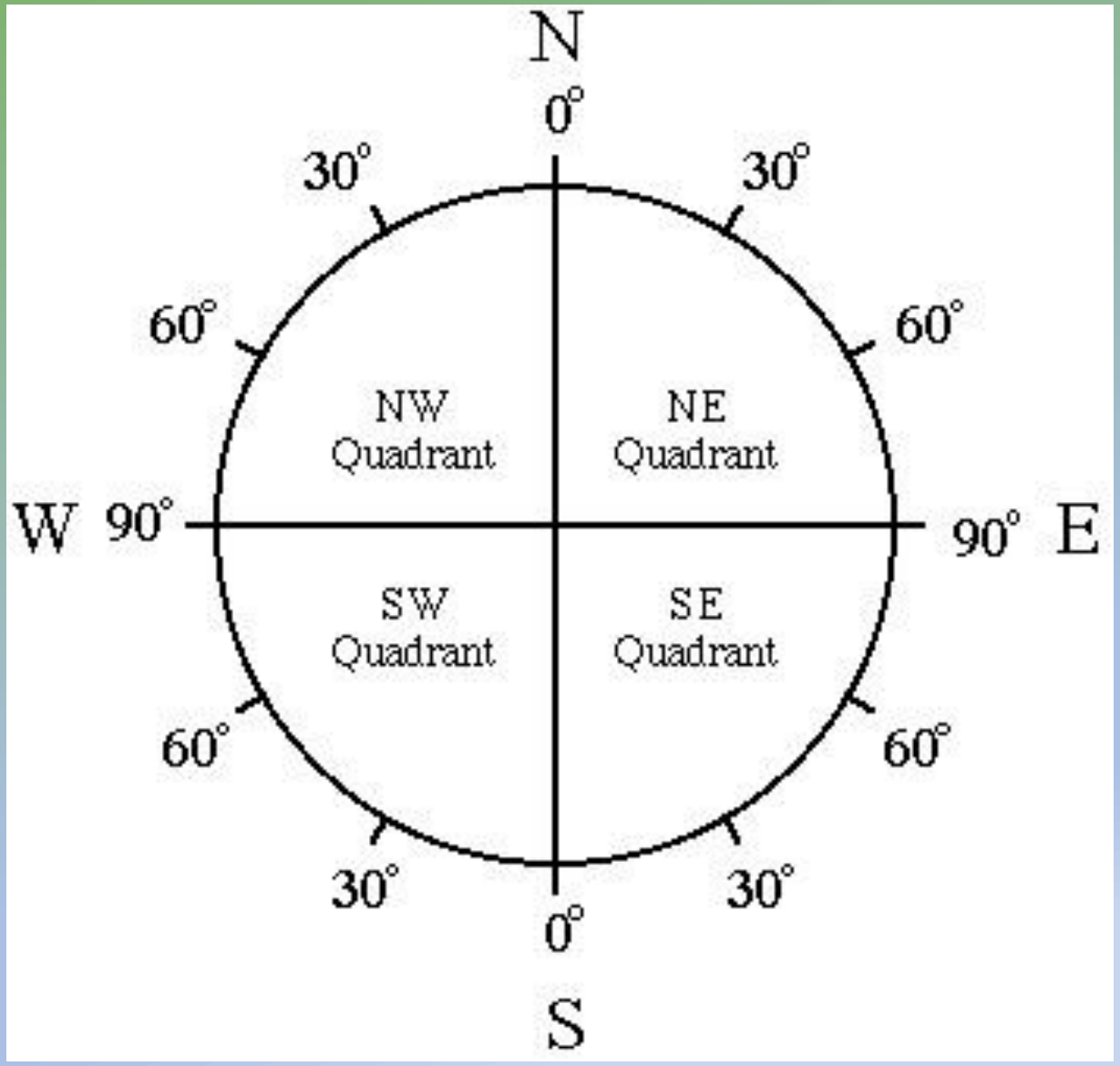
Latitude: 30.20833°

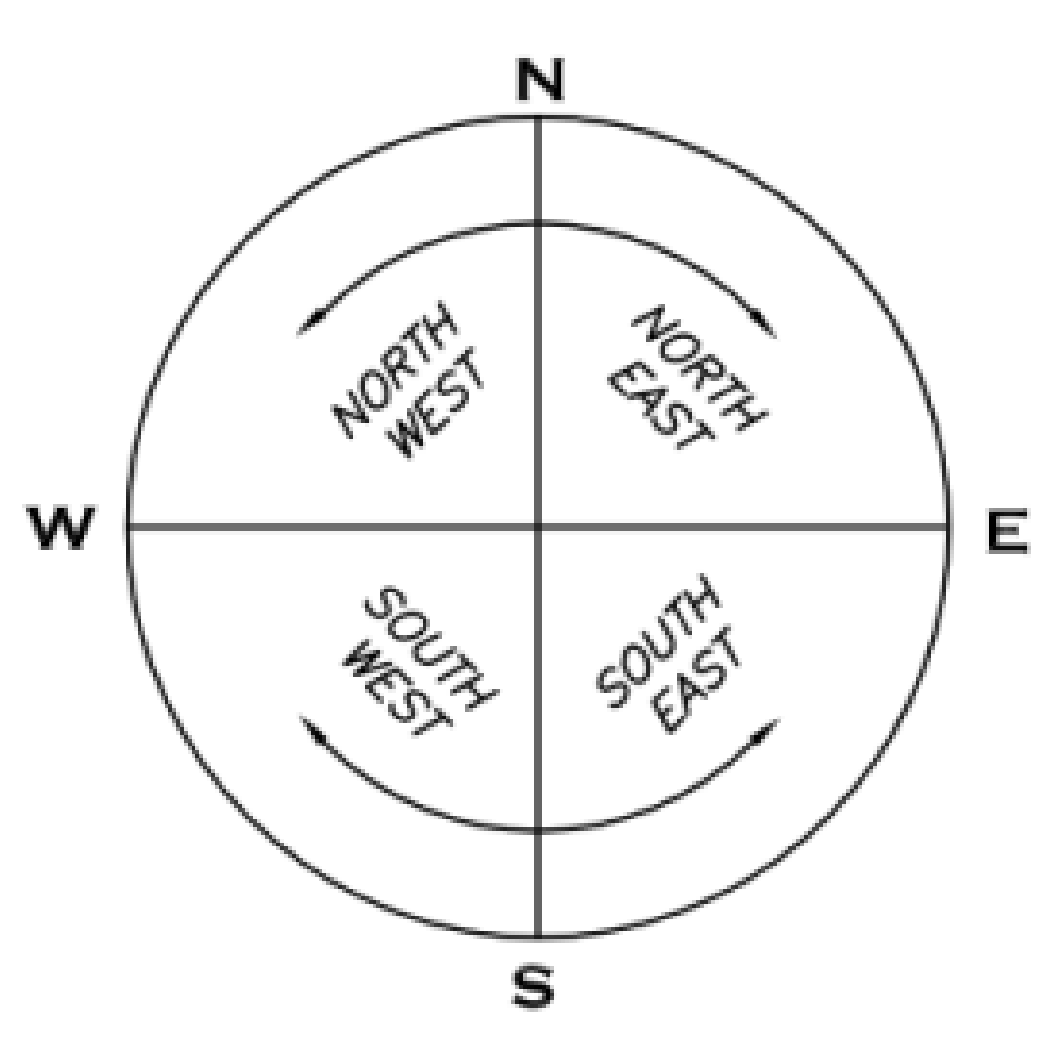
Longitute: -97.66583°

a minute is 1/60th of a degree (12/60=.2)
a second is 1/60th of minute
a second is 1/3600th of a degree (30/3600=.00833)
together = .20833

What are Metes: Quadrants

Definition:
each of four quarters of a circle



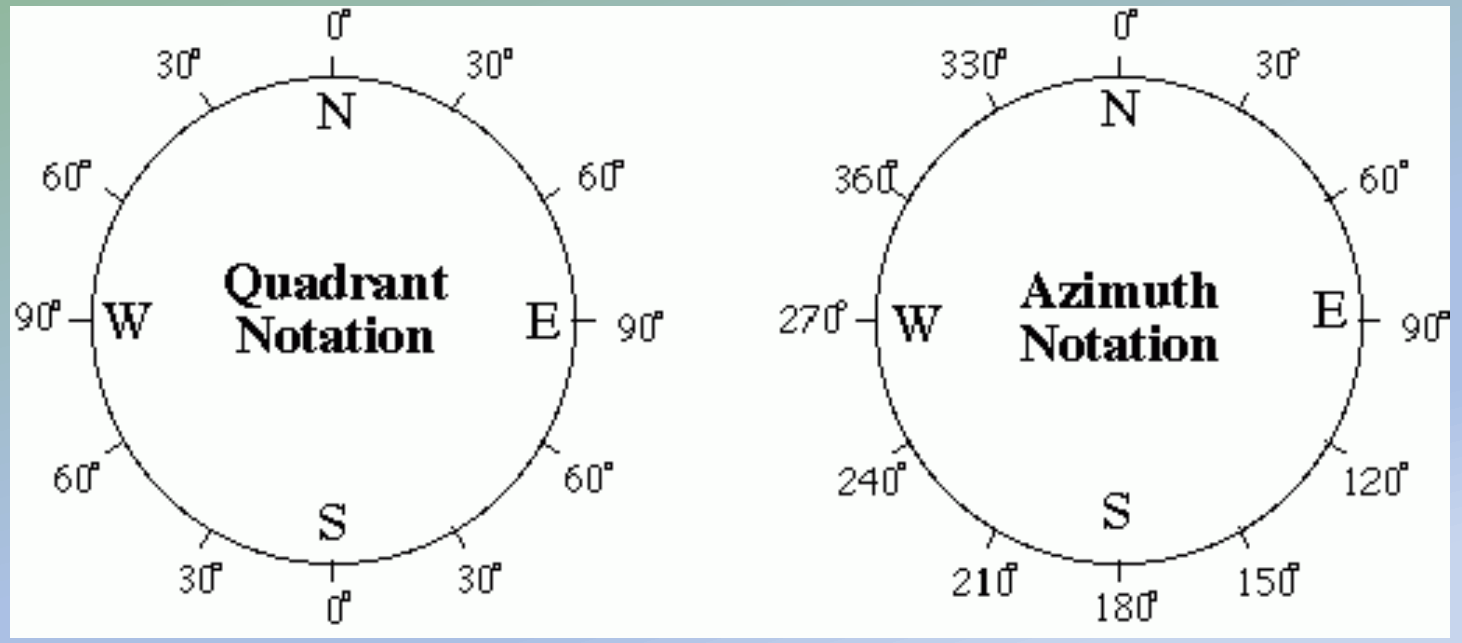


What are Metes: Quadrants

- Bearings are measured as the angle from due north and due south.
- They are stated as which quadrant they are in, as identified by the first and last letter of each bearing.

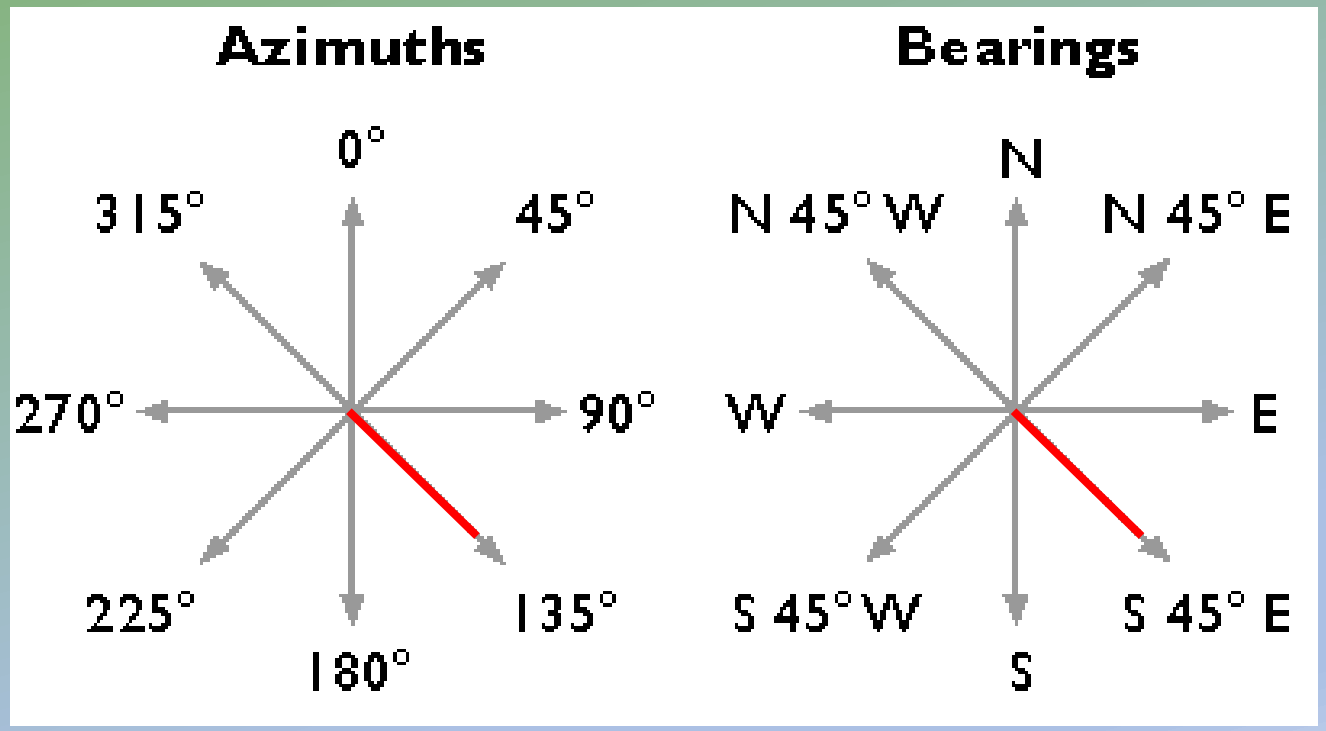
What are Metes: Quadrants

- A **bearing** is an angle less than 90° within a quadrant defined by the cardinal directions.
- An **azimuth** is an angle between 0° and 360° measured clockwise from North.

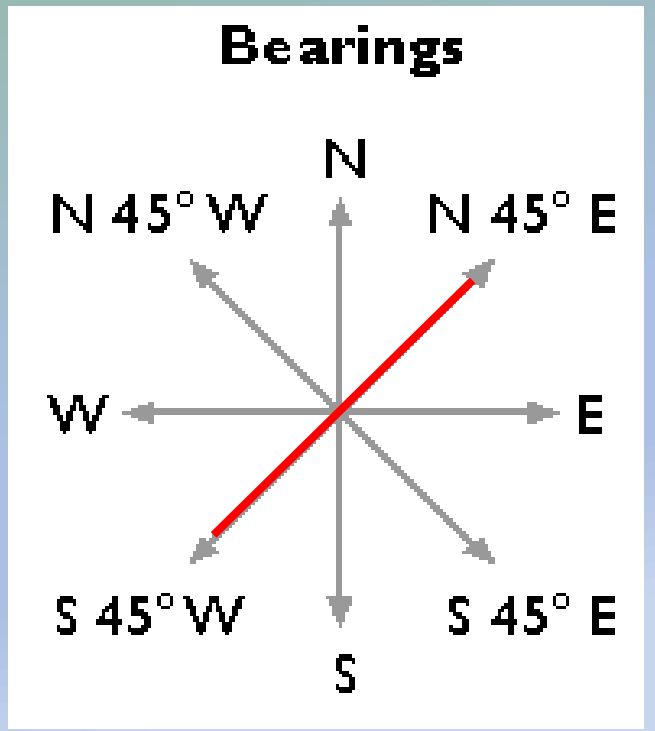


What are Metes: Bearings

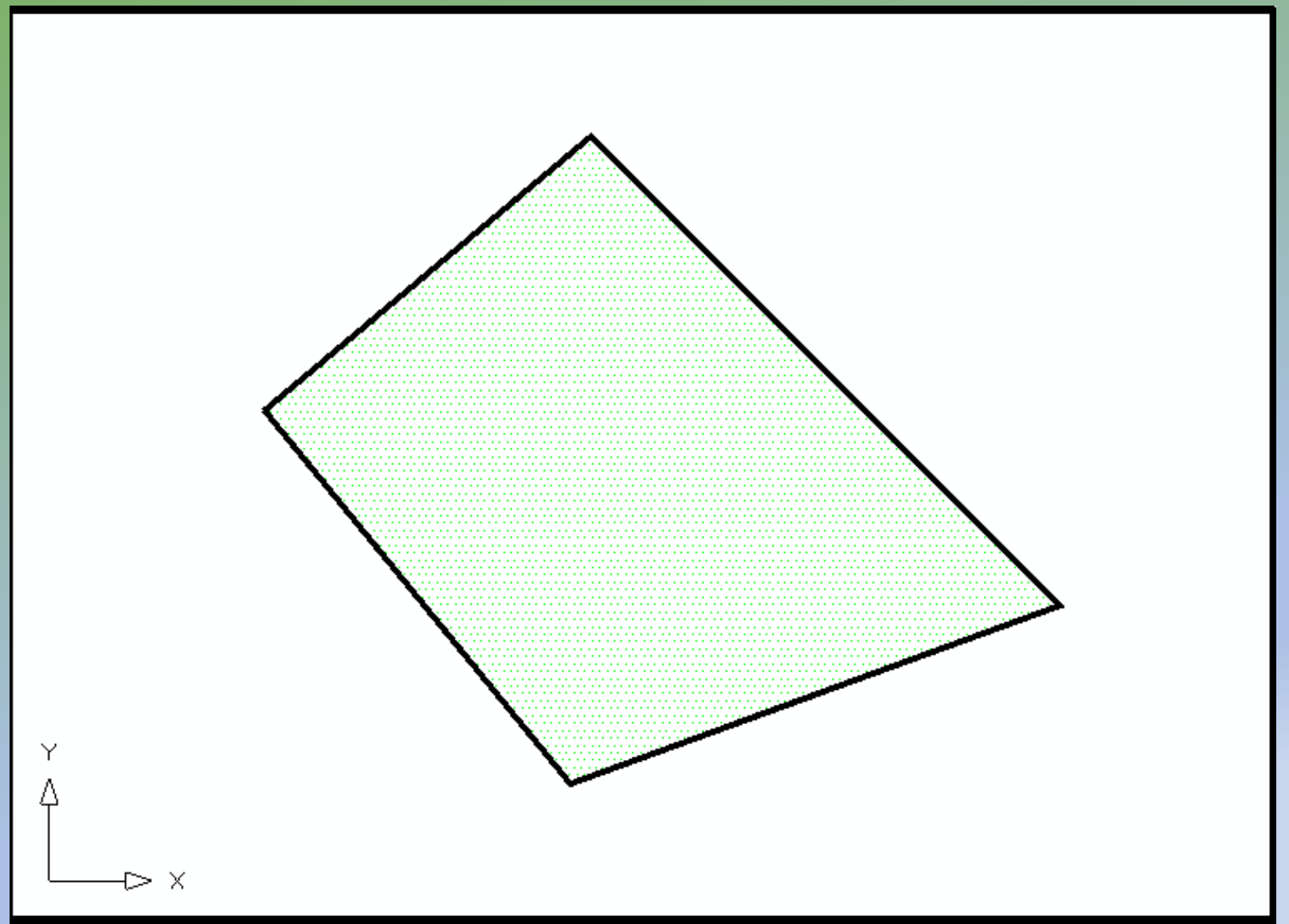
"135°“ and "South 45° East" are the same direction expressed as a bearing and as an azimuth.

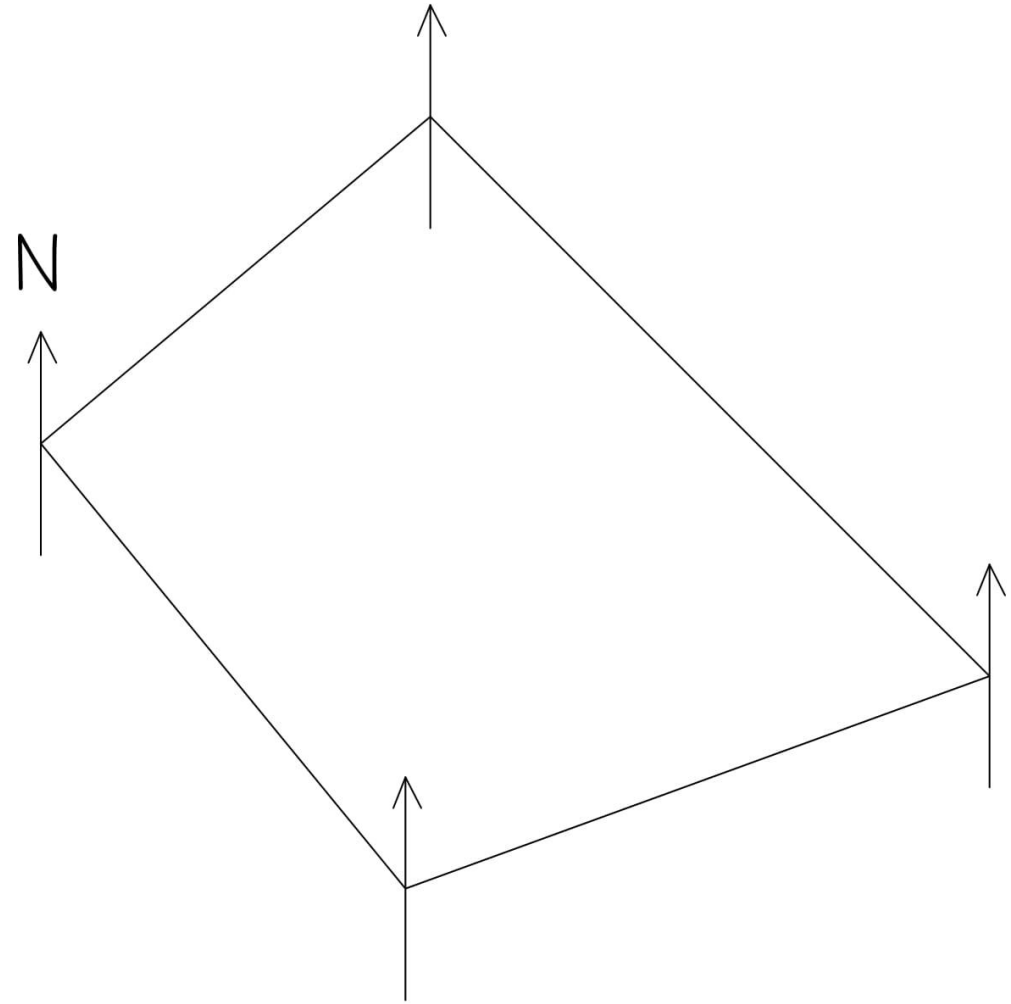


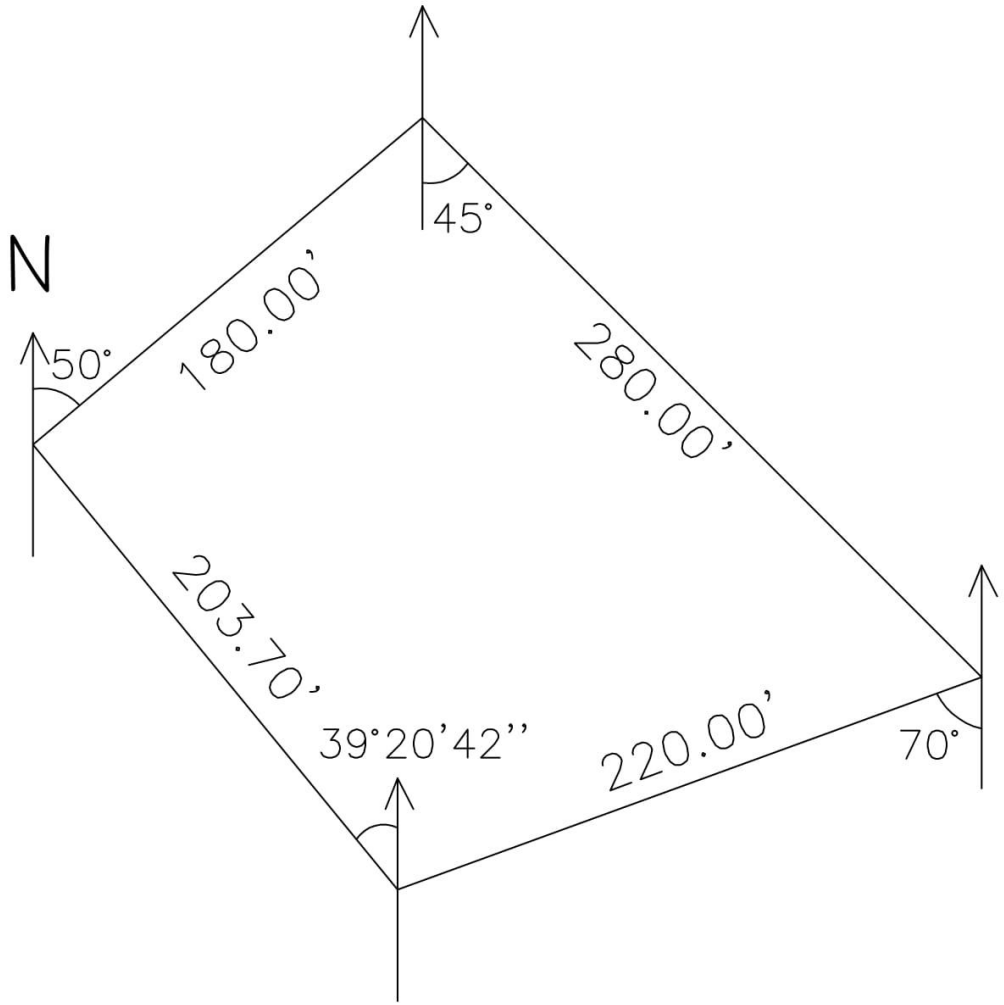
“North 45° East" and “South 45° West" are the same angle of a line.

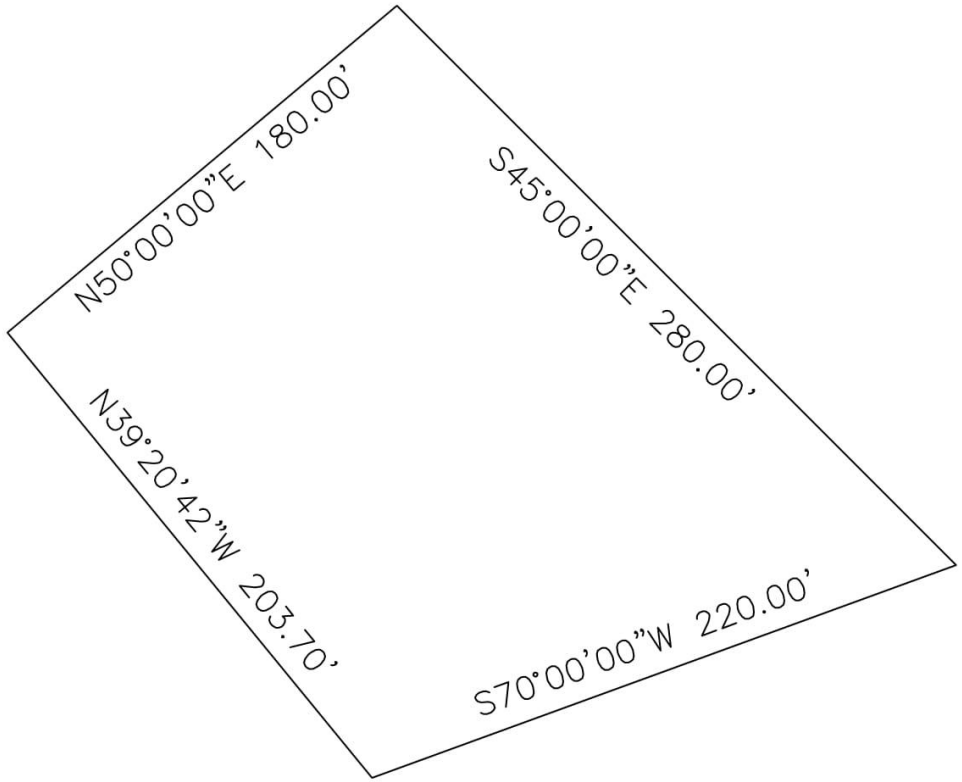


What are Metes:
EXAMPLE



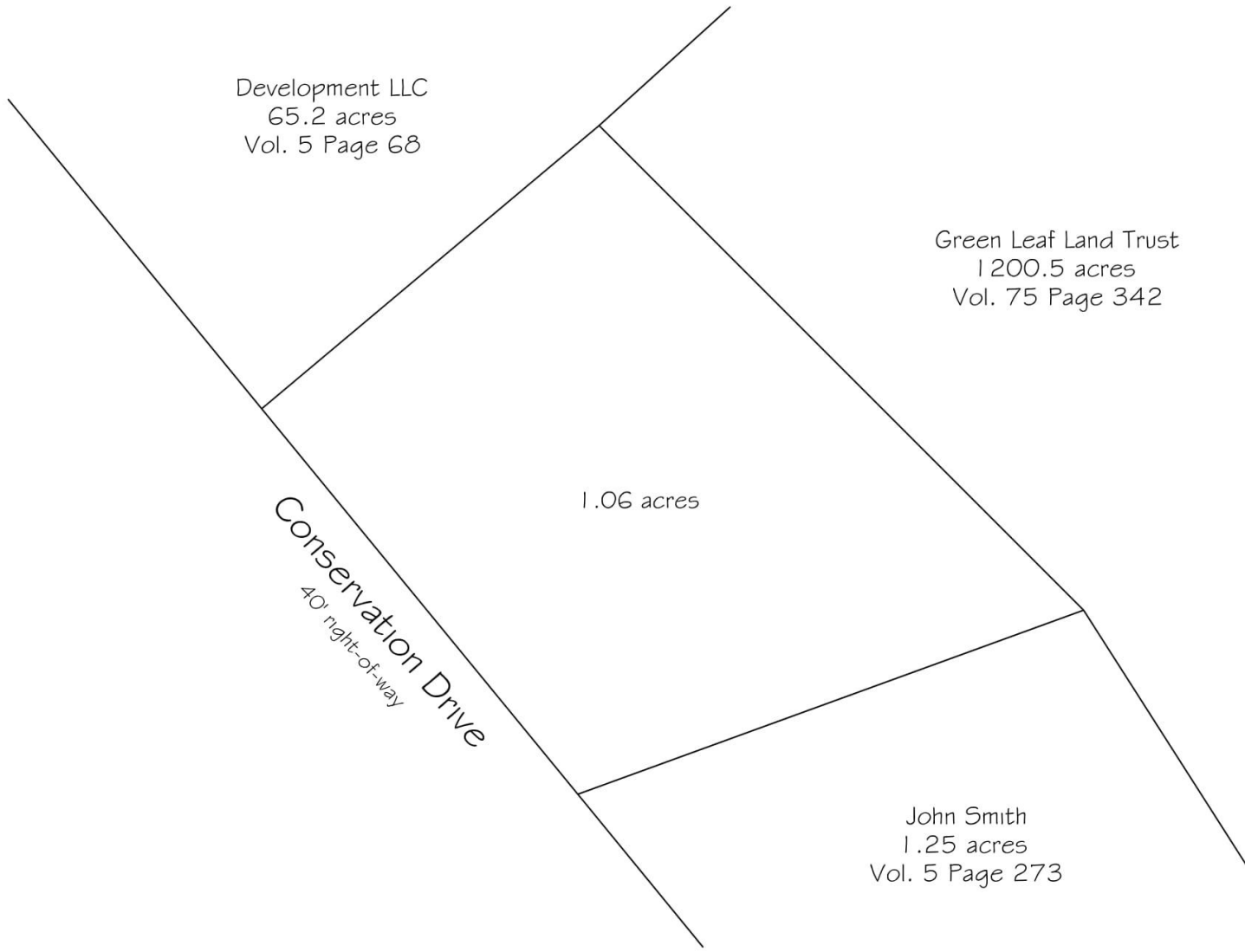

N


N
45°
50°
180.00'
280.00'
203.70'
39°20'42''
220.00'
70°

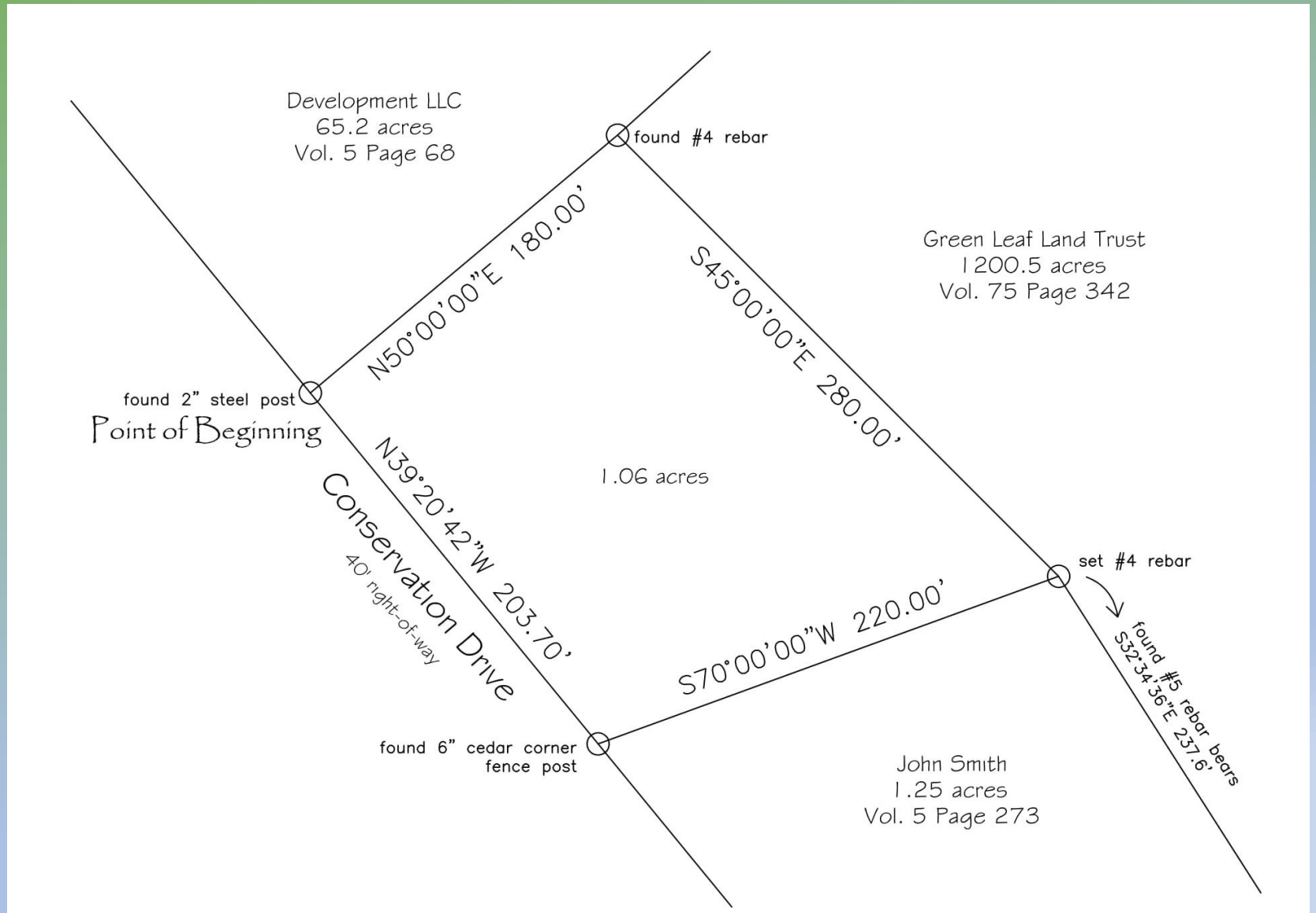

N50°00’00”E 180.00’
S45°00’00”E 280.00’
N39°20’42”W 203.70’
S70°00’00”W 220.00’

What are Bounds:

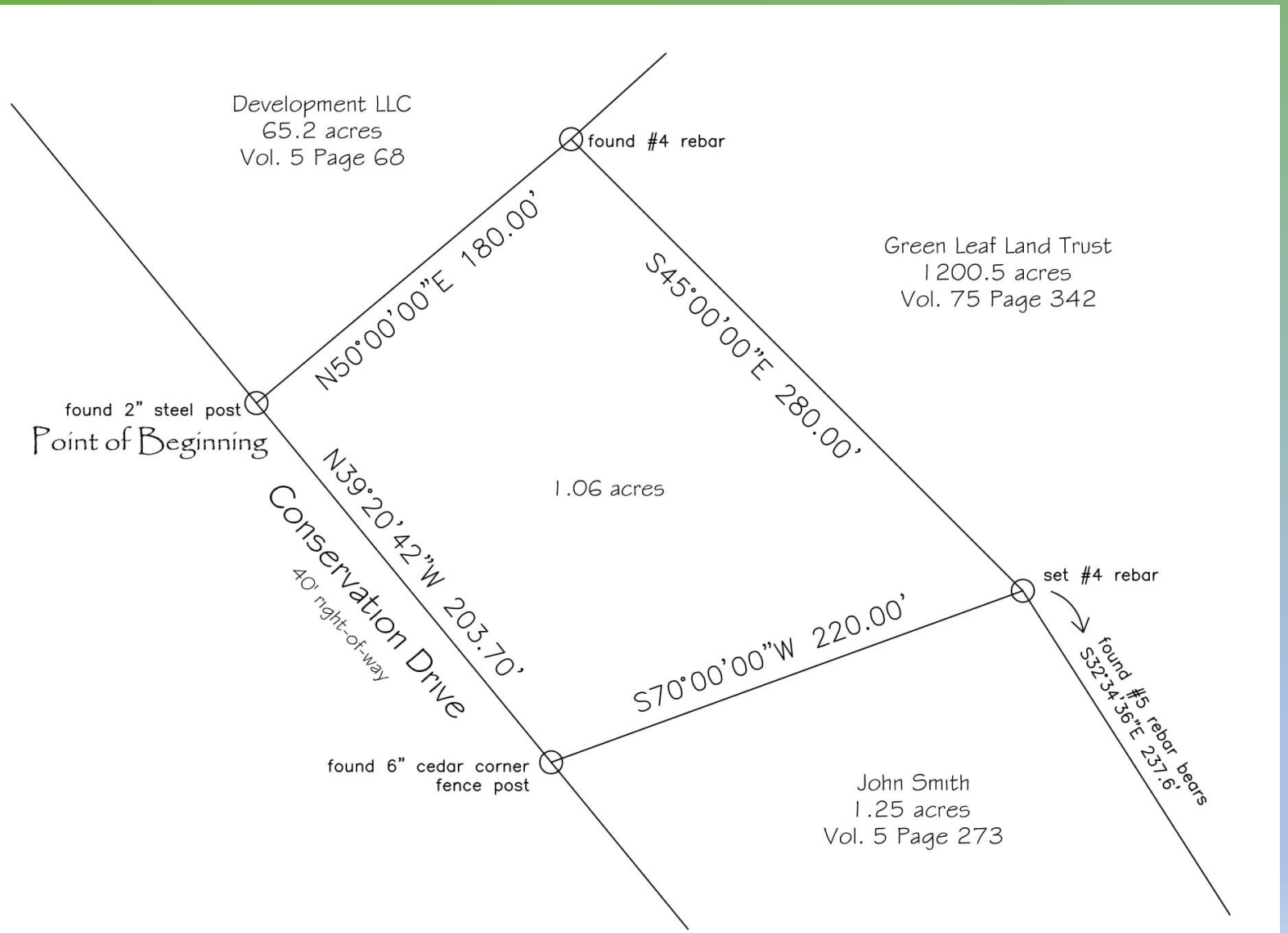
- Simple past tense and past participle of **bind**.
- Description by calls for adjoinders.
  - Adjoining - next to or joined with.



Monuments:
physical evidence of the metes and bounds



Metes and Bounds Description:



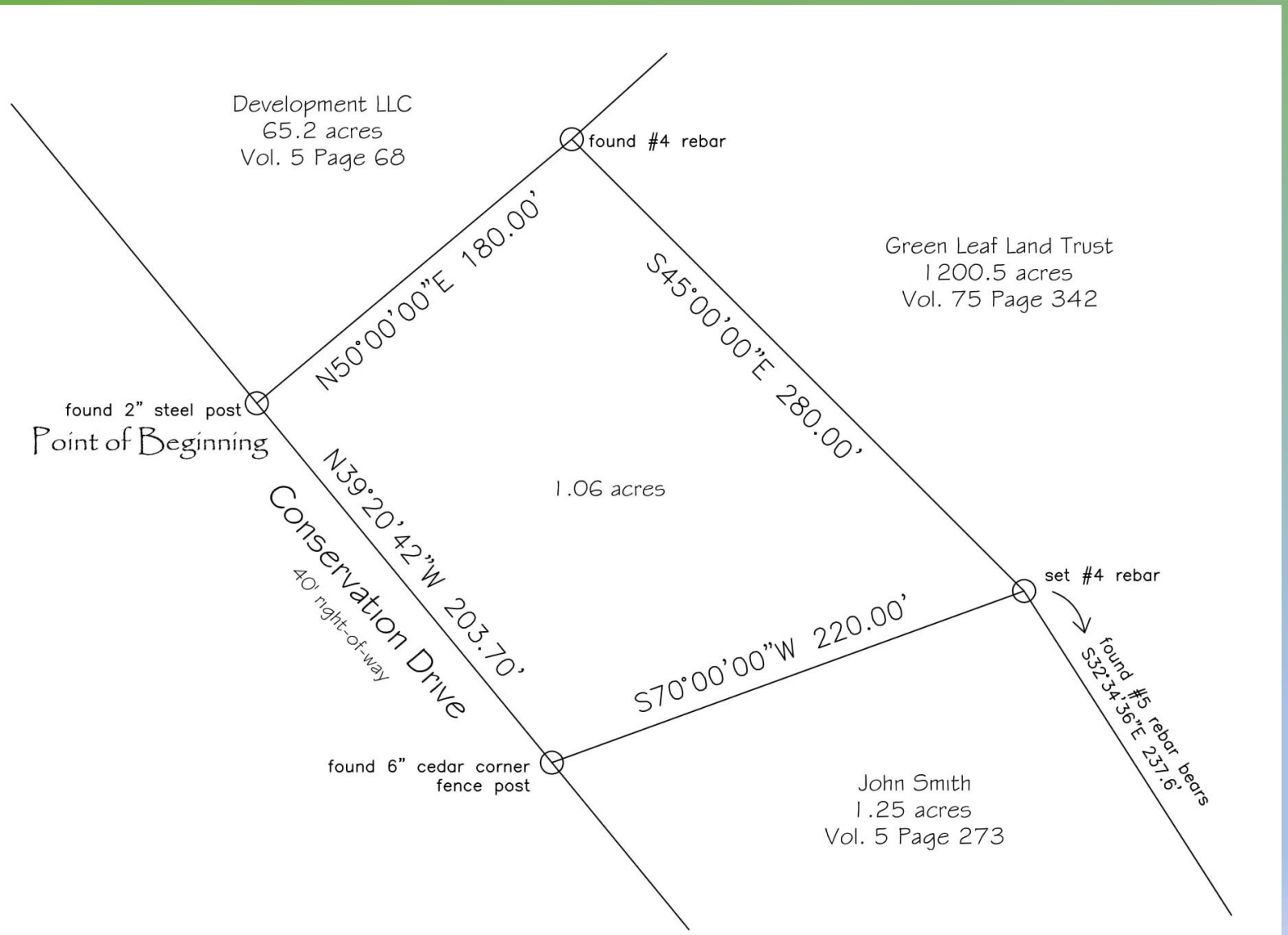
Title

General Description

Particular Description

Exceptions/Reservations

Certification

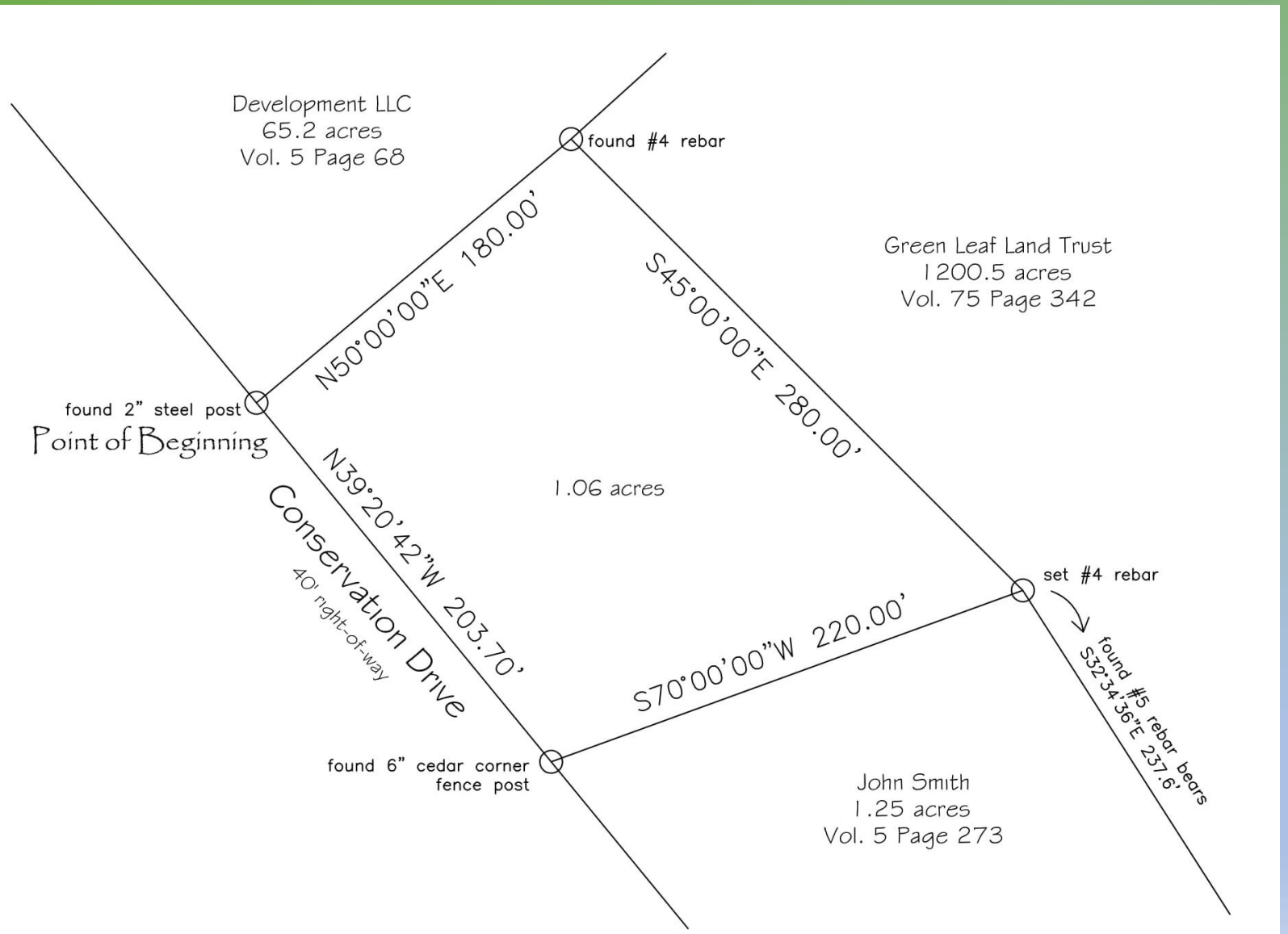


Title

Metes and Bounds Description of

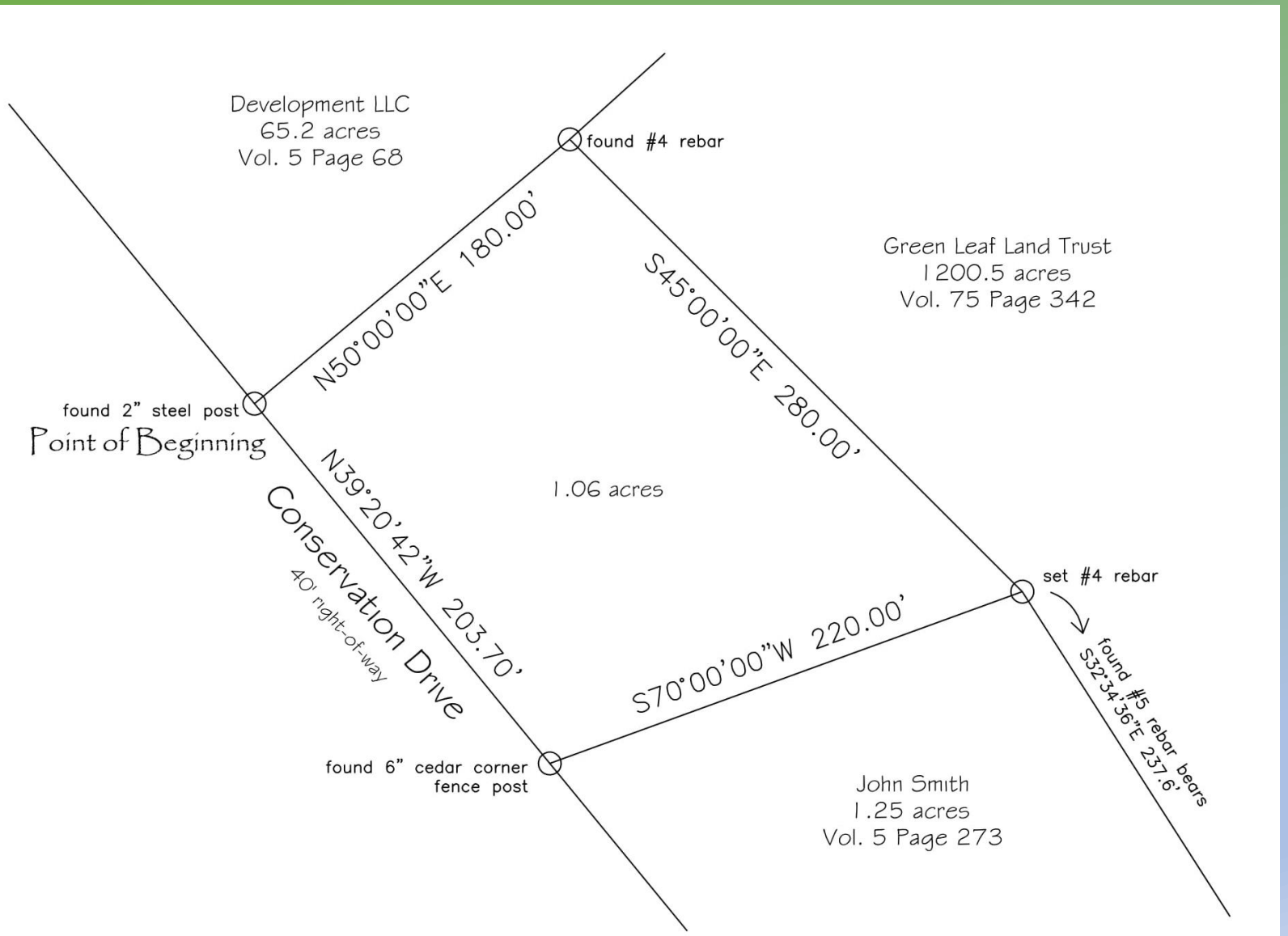
General Description:

1.06 acres as described as 1.05 acres in Volume 92 on Page 344 of the Official Public Records of Medina County, Texas, being more particularly described as:



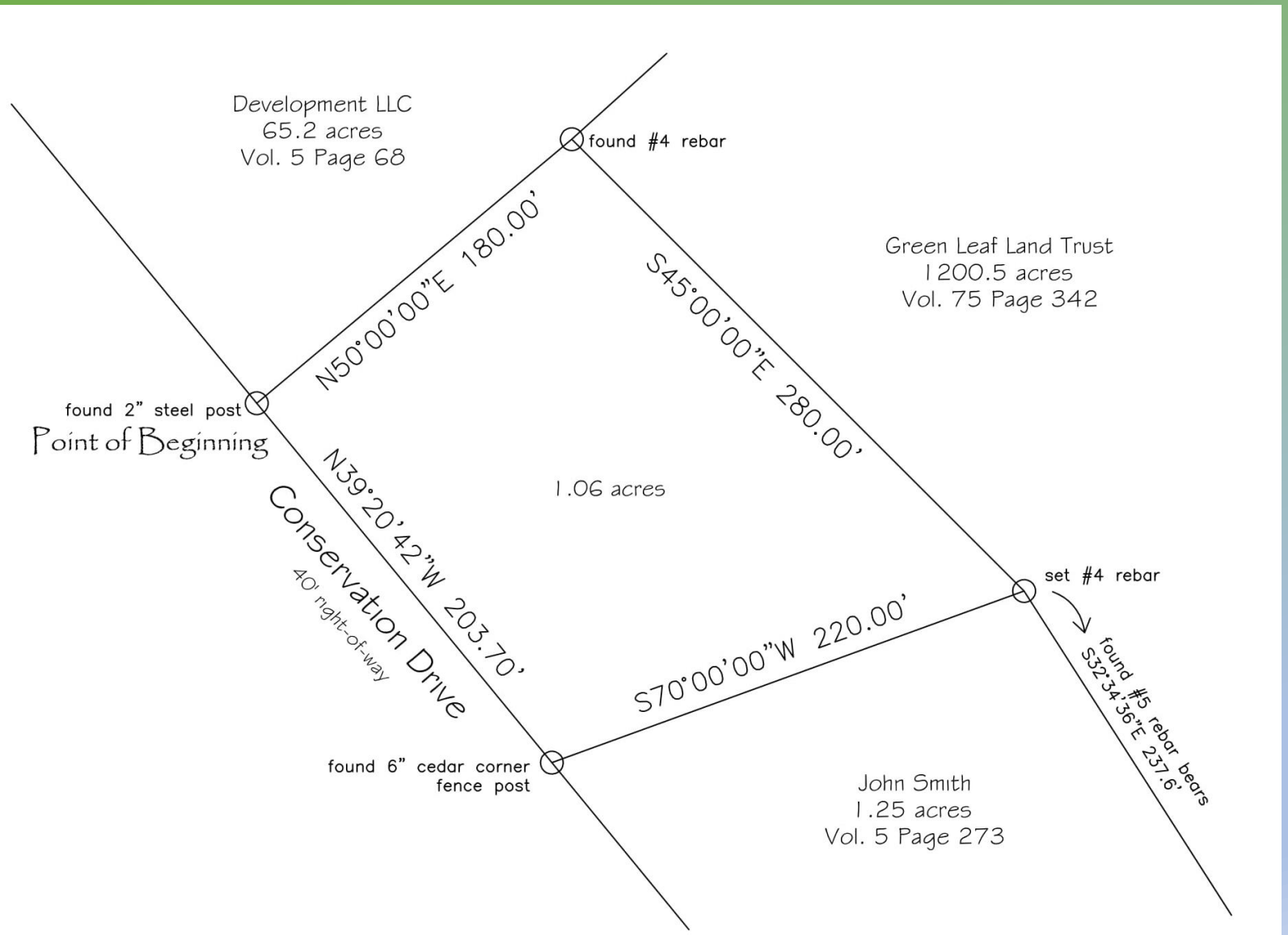
Particular Description:

BEGINNING at a found 2" steel post, the west corner of this tract, the south corner of a 65.2 acre tract, and on the east right-of-way line of Conservation Drive;



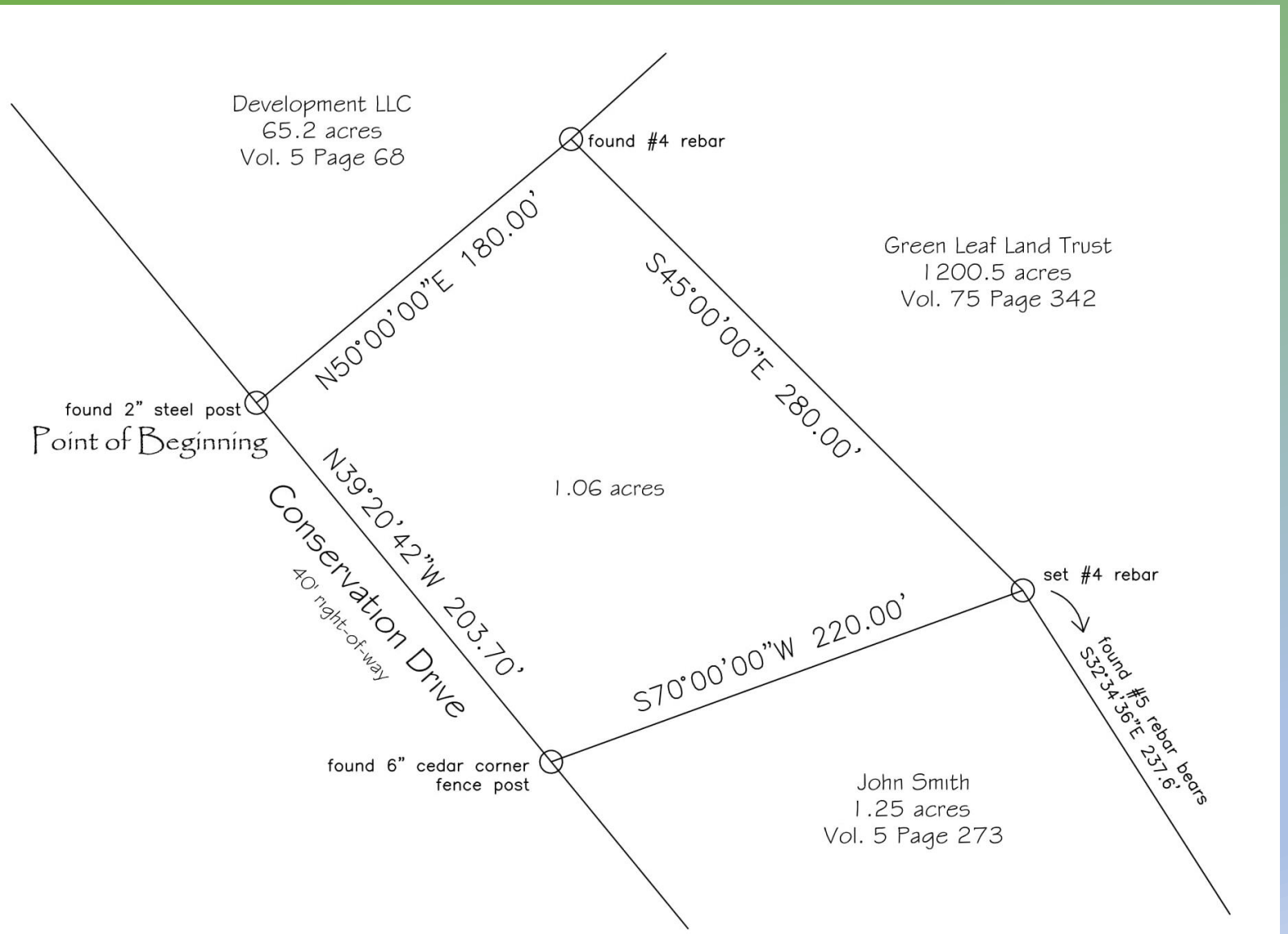
…cont. Particular Description:

THENCE N50°00'00"E 180.00’ with the south line of the 65.2 acre tract to a found #4 rebar, the north corner of this tract, and the west corner of a 1200.5 acre tract described in Volume 75 on Page 342, on the southeast line of the 65.2 acre tract;



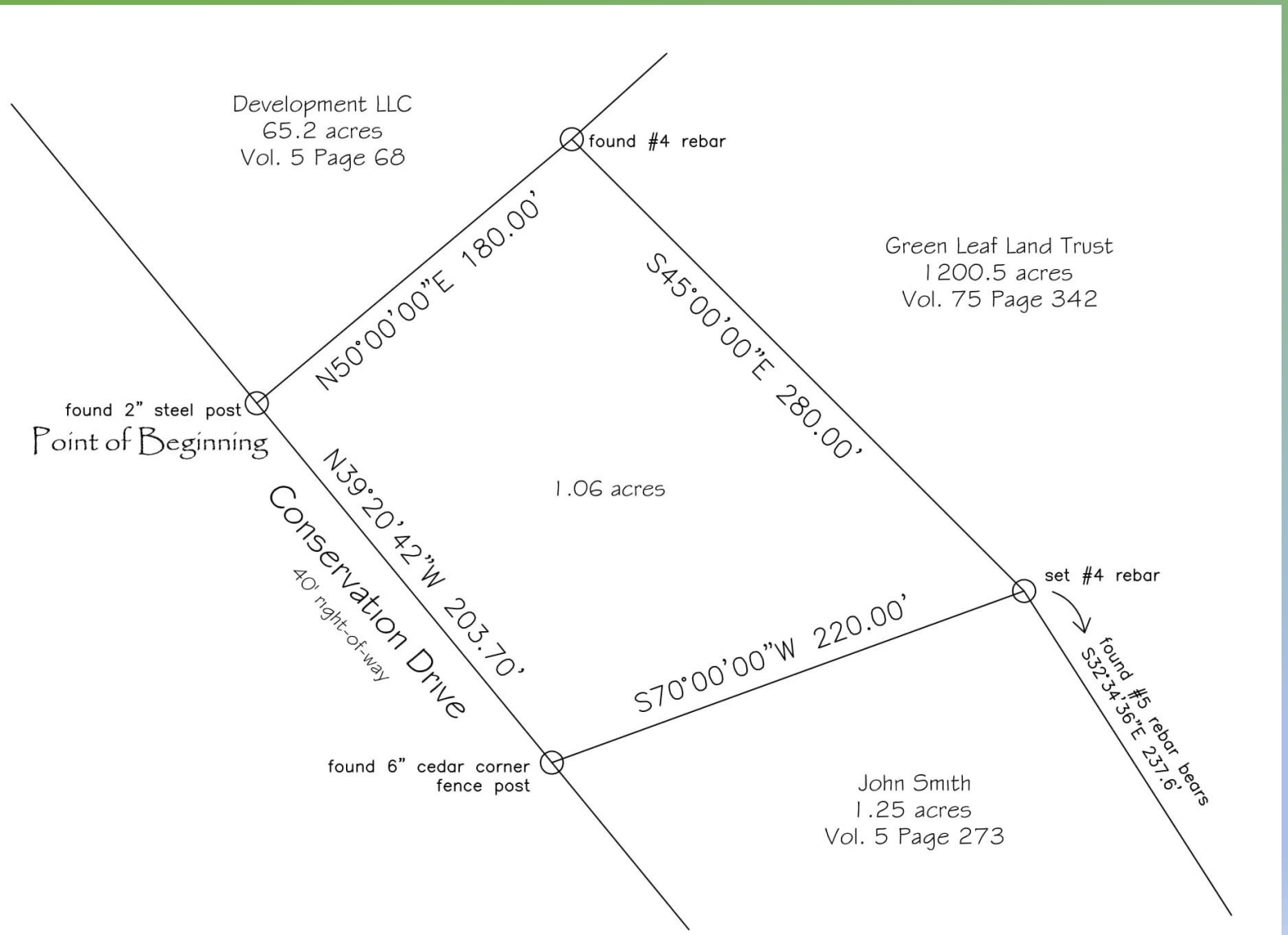
...cont. Particular Description:

THENCE S45°00'00"E 280.00' with the west line of the 1200.5 acre tract to a set #4 rebar, the east corner of this tract, and the north corner of a 1.25 acre tract described in Vol. 5 on Page 273 of the Official Public Records, on the west line of the 1200.5 acre tract, and from which a found #5 rebar, the southeast corner of the 1.25 acre tract, bears S32°34'36"E 237.6';



…cont. Particular Description:

THENCE S70°00'00"W 220.00'
with the north line of the 1.25 acre tract to a found 6" cedar corner fence post, the south corner of this tract, and the west corner of the 1.25 acre tract, on the east right-of-way line of Conservation Drive;



...cont. **Particular Description:**

THENCE N39°20'42"W 203.70’ with the east right-of-way line of Conservation Drive to the POINT OF BEGINNING, containing 1.06 acres of land.

Exceptions and Reservations

Certification

DISCLAIMER

Solution:
Equations Used

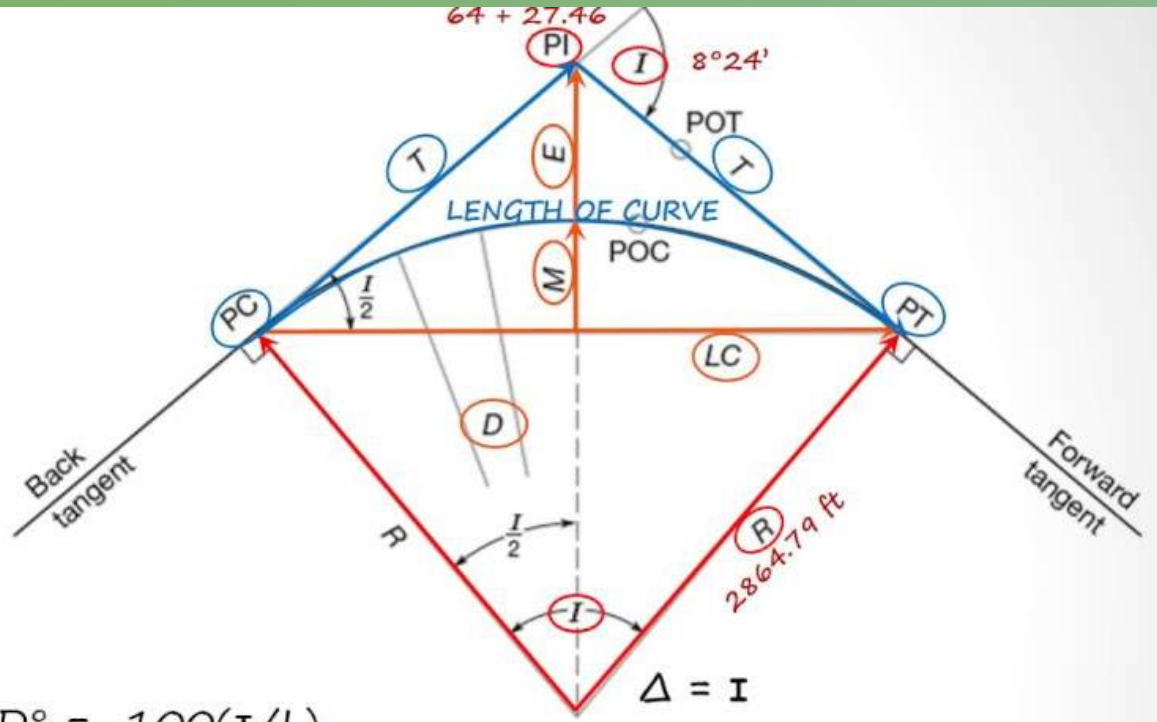
L = length of curve (arc)
Δ or I = central angle
LC = long chord
D° = Degree of Curvature
R = radius
T = tangent length
PI = intersection of tangents
E = external distance
M = middle ordinate distance

$L = 2\pi R\,(I / 360°)$

$T = R \tan(I/2)$

PC Station = PI − T

$LC = 2R \sin(I/2)$



$D° = 100(I/L)$

(note: I in degrees)

PT Station = PC + L

$M = R(1 - \cos(I/2))$

$$E = R\left(\frac{1}{\cos\left(\frac{I}{2}\right)} - 1\right)$$

Curves

Enter chord and chord bearing to close the perimeter and adjust to fit later

Make a Map

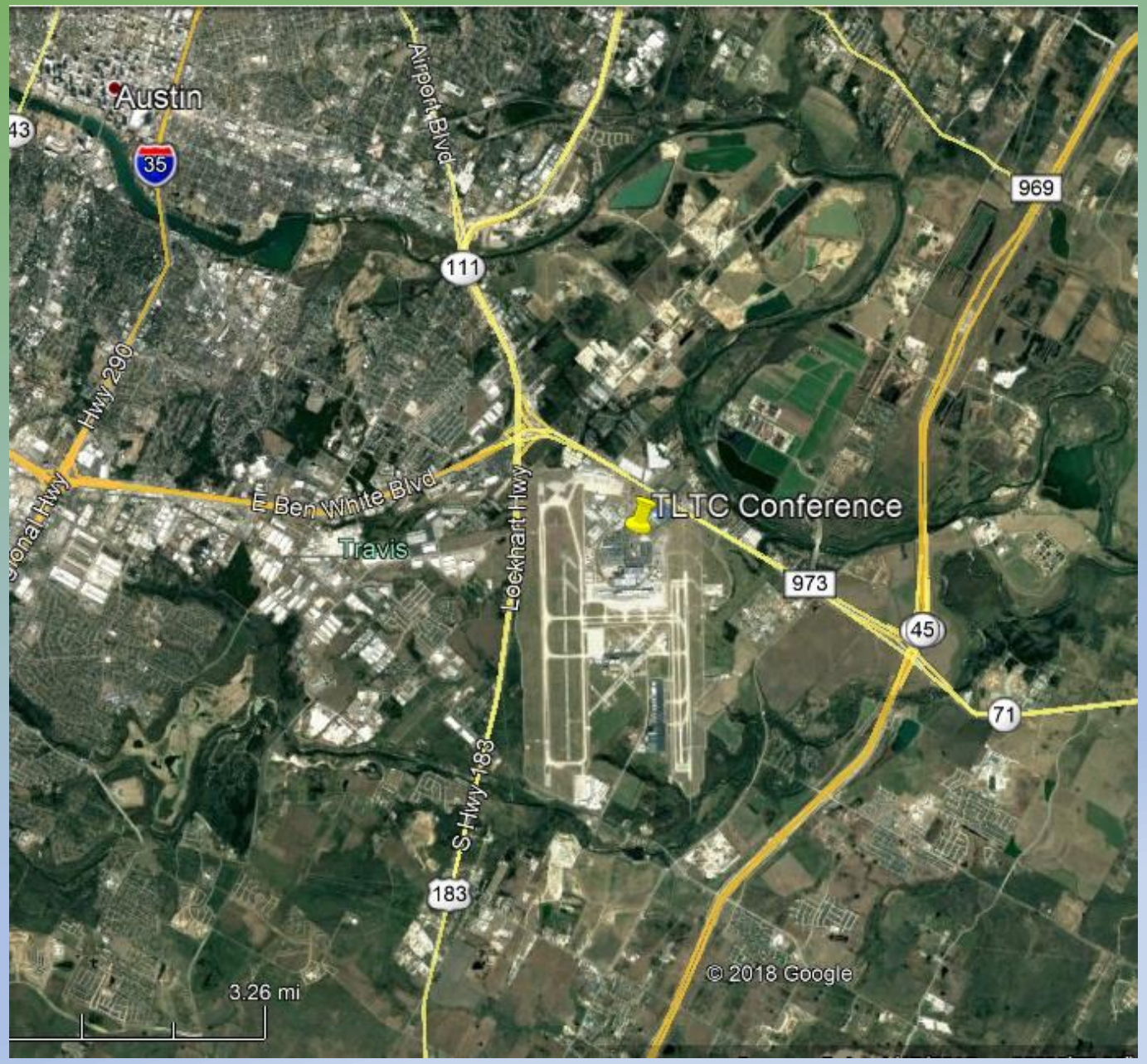
Get the data:

Google Earth or GIS w/ basemap

County

Approximate boundaries –

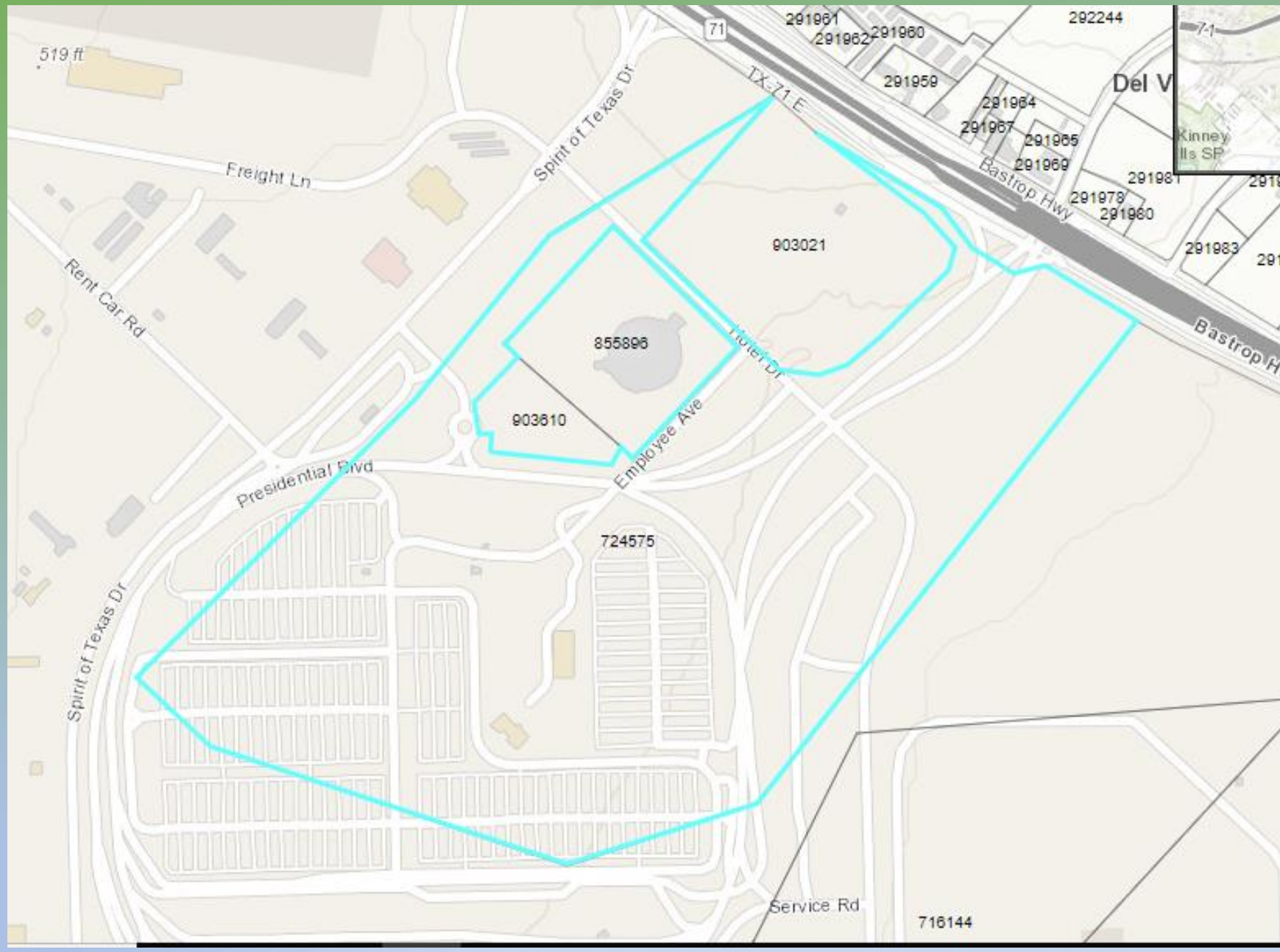
create a KMZ



Make a Map

Get the data:

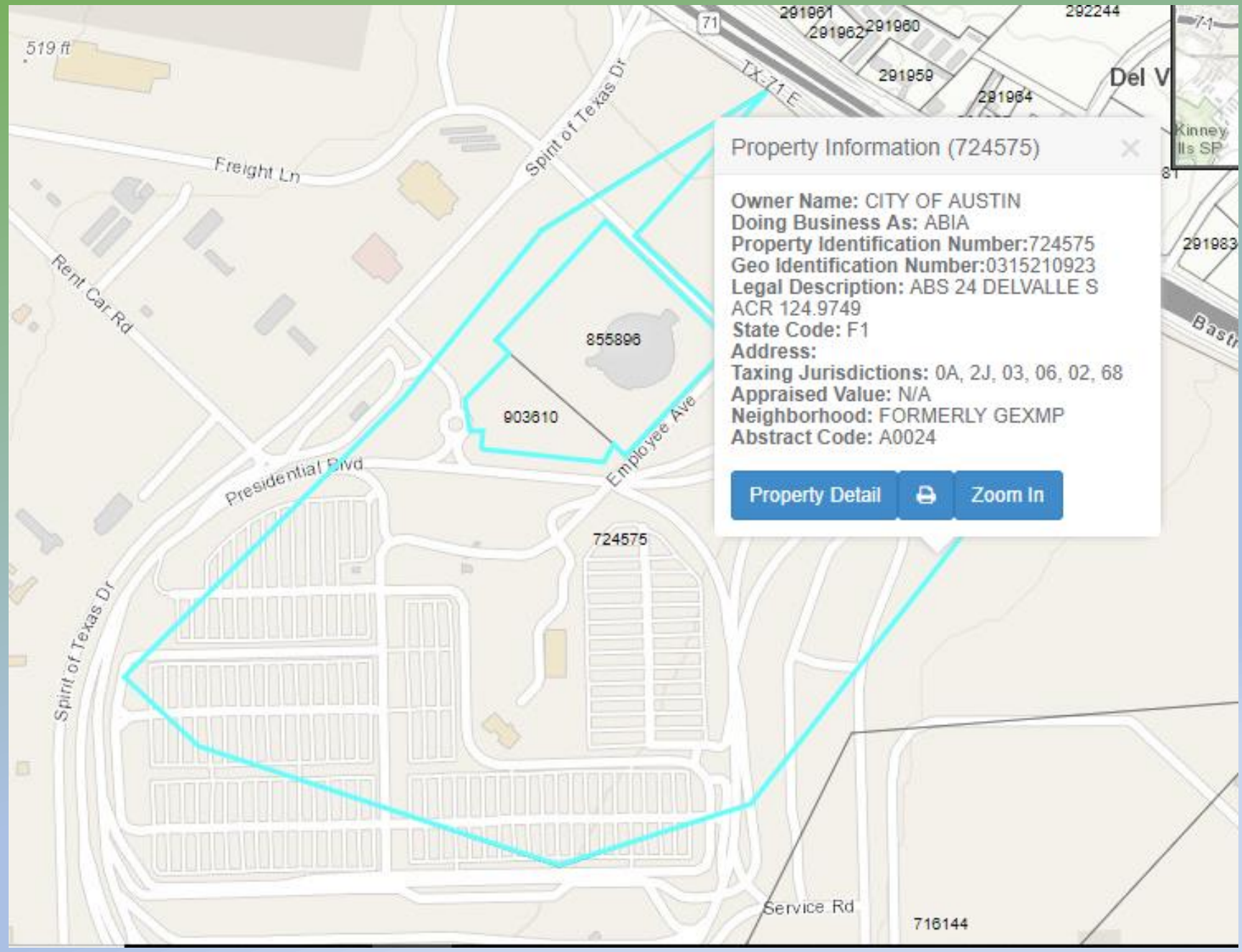
County Appraisal District



Make a Map

Get the data:

County Appraisal District



Owner

Name:	CITY OF AUSTIN	Owner ID:	571690
Mailing Address:	PO BOX 1088 AUSTIN , TX 78767-1088	% Ownership:	100.0000000000%
		Exemptions:	EX-XV

▸ Values

▸ Taxing Jurisdiction

▸ Improvement / Building

▸ Land

▸ Roll Value History

▾ Deed History - (Last 3 Deed Transactions)

#	Deed Date	Type	Description	Grantor	Grantee	Volume	Page	Deed Number
1	8/10/2005	DW	DEED WITHOUT WARRANTY	U S GOVERNMENT	CITY OF AUSTIN			2005160431TR

Questions Please Call (512) 834-9317

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Travis County
County Clerk Web Search

Welcome Visitor.

Welcome | Marriage Records | Official Public Records | FAQ

OPR Document Access

Permanent Index From 01/01/1988 to 02/01/2019
Temporary Index from 02/02/2019 to 02/09/2019
Images from 04/05/1999

Search | Clear Form

General

Grantor:

Grantee: Lastname Firstname

Date Filed From: mm/dd/yyyy **To:** mm/dd/yyyy

Instrument # From: 2005160431TR **To:**

Book: **Page:**

Document Type:

- ABANDONMENT OF ASSUMED NAMES
- ABSTRACT OF JUDGMENT
- AFFIDAVIT
- AFFIDAVIT OF CLAIM
- AFFIDAVIT OF HEIRSHIP

DEED 2005160431
23 PGS

DEED WITHOUT WARRANTY

FORMER BERGSTROM AIR FORCE BASE

TRAVIS COUNTY, TEXAS

NOTICE OF CONFIDENTIALITY RIGHTS

IF YOU ARE A NATURAL PERSON, YOU MAY REMOVE OR STRIKE ANY OR ALL OF THE FOLLOWING INFORMATION FROM THIS INSTRUMENT BEFORE IT IS FILED FOR RECORD IN THE PUBLIC RECORDS: YOUR SOCIAL SECURITY NUMBER OR YOUR DRIVER'S LICENSE NUMBER.

I. PARTIES

This Deed made this 10th day of August, 2005, by and between the United States of America, acting by and through the Secretary of the Air Force whose address is Washington, D.C., under and pursuant to the Federal Property and Administrative Services Act of 1949, approved June 30, 1949, (63 Stat. 377), 40 U.S.C. § 101, et seq., as amended, and regulations and orders promulgated thereunder; the Defense Base Closure and Realignment Act of 1990, Pub. L. No. 101-510, as amended, and regulations and orders promulgated thereunder; and a delegation from the Administrator of General Services to the Secretary of Defense, and a subsequent delegation from the Secretary of Defense to the Secretary of the Air Force, party of the first part, as Grantor, and the City of Austin, Texas, a body politic created, operating, and existing under and by virtue of the laws of the State of Texas, party of the second part, as Grantee.

WITNESSETH THAT:

WHEREAS, the Grantor is the owner of the real property described herein, located within the former Bergstrom Air Force Base, situated in Travis County, Texas; and

WHEREAS, the Grantee provided to the United States the money to purchase the real property described herein, under the condition that the United States retain title until such property was abandoned as a permanent Air Base, at which time the Grantee could elect to require the Grantor to convey such land and the improvements thereon to the Grantee; and

WHEREAS, the real property described herein was duly declared surplus and available for disposal pursuant to the powers and authority contained in the provisions of the Defense Base Closure and Realignment Act of 1990, Pub. L. No. 101-510, as amended, and orders and regulations promulgated thereunder; and

MACIAS & ASSOCIATES, Inc.
LAND SURVEYORS

Exhibit "A"
(PARCEL F)

United States of America
to
City of Austin

LEGAL DESCRIPTION

OF A 157.876 ACRE PARCEL OF LAND OUT OF THE SANTIAGO DEL VALLE TEN LEAGUE GRANT, ABSTRACT NO. 24, TRAVIS COUNTY, TEXAS; ALSO BEING OUT OF 1.) TRACT NO. A-1, A 168.96 ACRE PARCEL CONVEYED TO THE UNITED STATES OF AMERICA BY INSTRUMENT OF RECORD IN VOLUME 709, PAGE 181,, DEED RECORDS OF SAID TRAVIS COUNTY; 2.) TRACT NO. A-2, COMPRISED OF TWO PARCELS 87.84 ACRES AND 295.59 ACRES CONVEYED TO THE UNITED STATES OF AMERICA BY INSTRUMENT OF RECORD IN VOLUME 708, PAGE 115, DEED RECORDS OF SAID TRAVIS COUNTY; 3.) TRACT NO. A-3, A 103.62 ACRE PARCEL CONVEYED TO THE UNITED STATES OF AMERICA BY INSTRUMENT OF RECORD IN VOLUME 709, PAGE 390, DEED RECORDS OF SAID TRAVIS COUNTY; 4.) TRACT NO. A-19, A 139.85 ACRE PARCEL CONVEYED TO THE UNITED STATES OF AMERICA BY INSTRUMENT OF RECORD IN VOLUME 709, PAGE 67, DEED RECORDS OF SAID TRAVIS COUNTY; AND A PORTION OF THE ABANDONED OLD BASTROP ROAD; SAID 157.876 ACRE PARCEL AS SHOWN ON THE ACCOMPANYING SKETCH, IS MORE PARTICULARLY DESCRIBED BY METES AND BOUNDS AS FOLLOWS:

BEGINNING at the westernmost corner of this parcel, being a point occupied by monitoring well MW8-15; also being SOUTH 2191.23 feet and EAST 2734.78 feet from a brass disk in concrete found in the centerline at the north end of the west runway 17R/35L;

THENCE, N 46° 16' 31" E, 1622.32 feet to a point occupied by monitoring well SM62MW at an angle point;

THENCE, N 40° 48' 20" E, 876.24 feet to a point occupied by monitoring well SM53MW at an angle point;

THENCE, N 59° 39' 54" E, 1093.78 feet to a point on the southwesterly right-of-way line of State Highway No. 71;

THENCE, the following six (6) courses with said right-of-way line:

1.) S 47° 23' 55" E, 234.00 feet to a TxDOT Type I concrete monument found at an angle point;

2.) S 58° 11' 43" E, 600.18 feet to a TxDOT Type I concrete monument found at an angle point;

3.) S 35° 21' 42" E, 206.16 feet to a ½" iron rod set at an angle point;

4.) S 58° 11' 43" E, 210.00 feet to a ½" iron rod set at an angle point;

5.) N 76° 48' 17" E, 127.28 feet to a ½" iron rod set at an angle point;

6.) S 58° 17' 05" E, 442.41 feet to a point at the easternmost corner of this parcel, from said point, a TxDOT Type I concrete monument found bears, S 58° 17' 05" E, 167.32 feet;

THENCE, S 39° 29' 51" W, 2538.95 feet to a point occupied by monitoring well SM08MW at an angle point;

THENCE, S 73° 33' 09" W, 823.14 feet to a point occupied by monitoring well SM14MW at an angle point;

THENCE, N 70° 30' 20" W, 1557.44 feet to a point occupied by monitoring well SM80MW at an angle point;

THENCE, N 45° 25' 30" W, 416.78 feet to the **POINT OF BEGINNING**, containing 157.876 acres of land.

BEARING BASIS NOTE

The bearings described herein are based on the Bergstrom Airport Coordinate System (BACS). The reference for this project is two brass disks in concrete found on the centerline of the west runway at the Austin – Bergstrom International Airport at an assumed bearing of NORTH.

THE STATE OF TEXAS §

KNOW ALL MEN BY THESE PRESENTS:

COUNTY OF TRAVIS §

That I, Henry A. Dufeau, a Registered Professional Land Surveyor, do hereby certify that the above description is true and correct to the best of my knowledge and belief and that the property described herein was determined by a survey made on the ground under my direction and supervision.

WITNESS MY HAND AND SEAL at Austin, Travis County, Texas this the 1st day of March, 2006.



MACIAS & ASSOCIATES, INC.
5410 South 1st Street
Austin, Texas 78745-3040

Henry A. Dufeau
Registered Professional Land Surveyor
No. 1707-State of Texas

REFERENCES

TCAD # 0315310601

Data Entry

- Quadrants are numbered 1-4
- Data entry starts with quadrant #
- Followed by the degrees, minutes, seconds, and distance

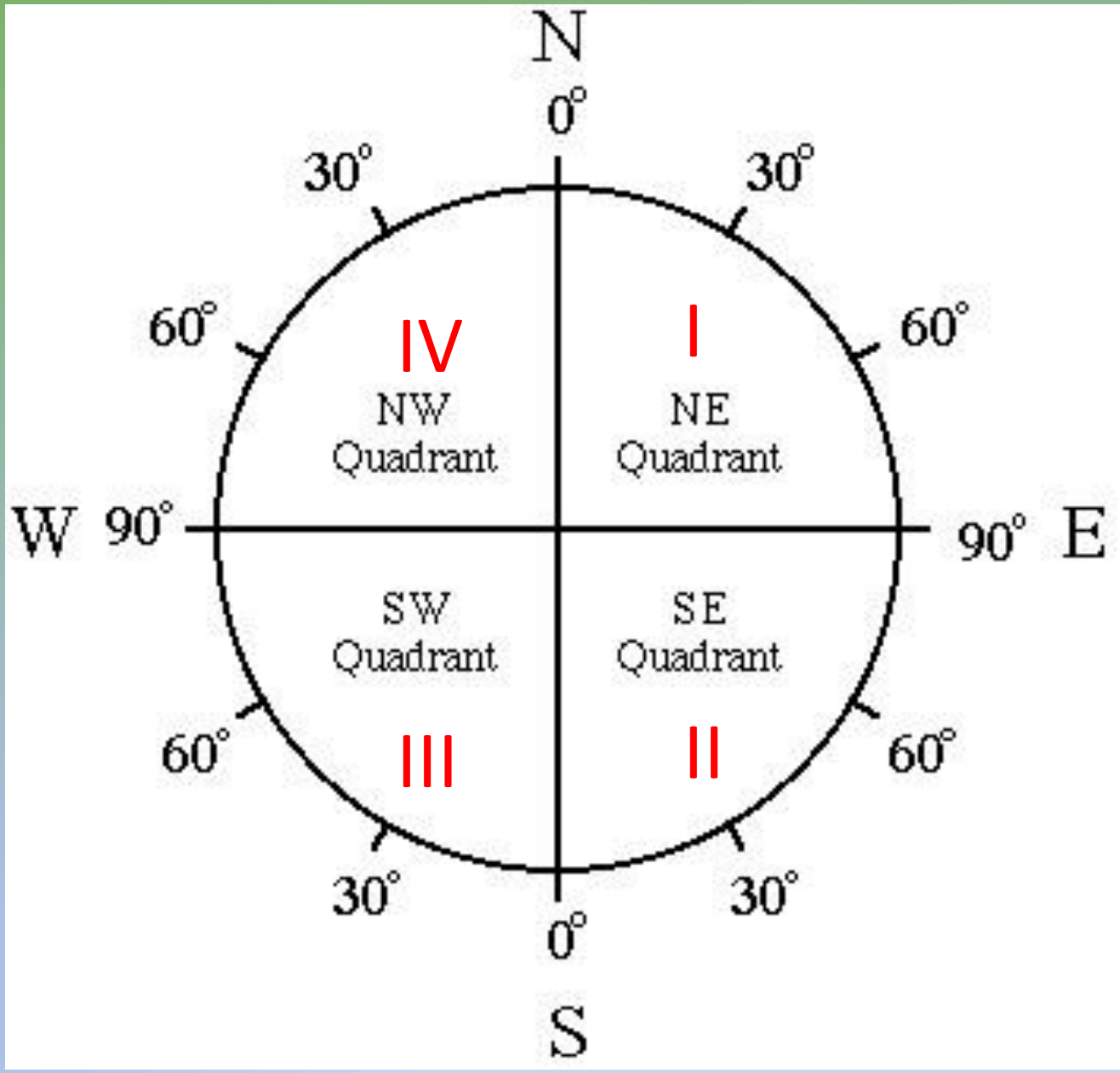
EXAMPLE

N46°16’31"E 1622.32’

Quadrant = 1

Bearing = 46.1631

Distance = 1622.32



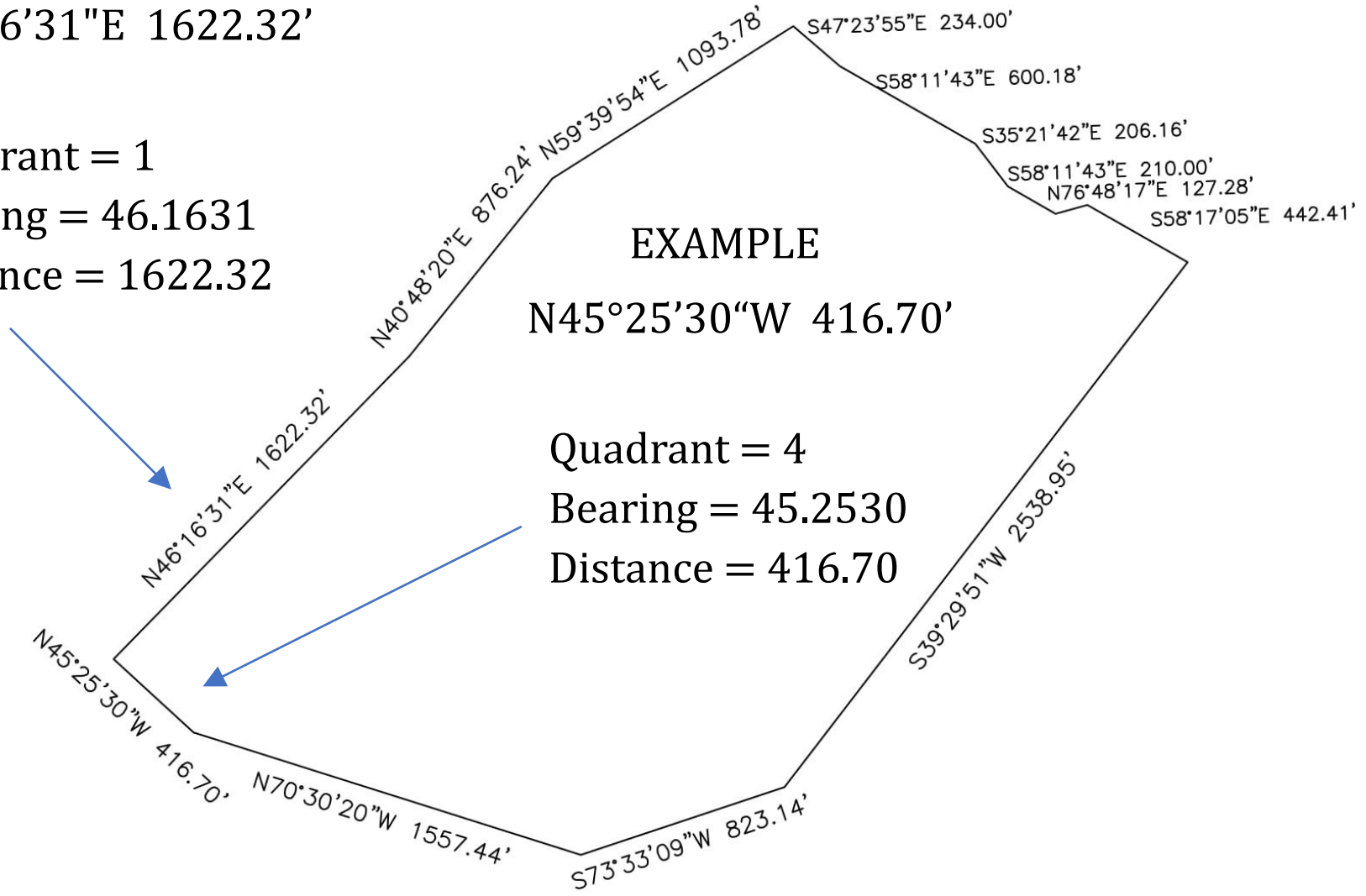
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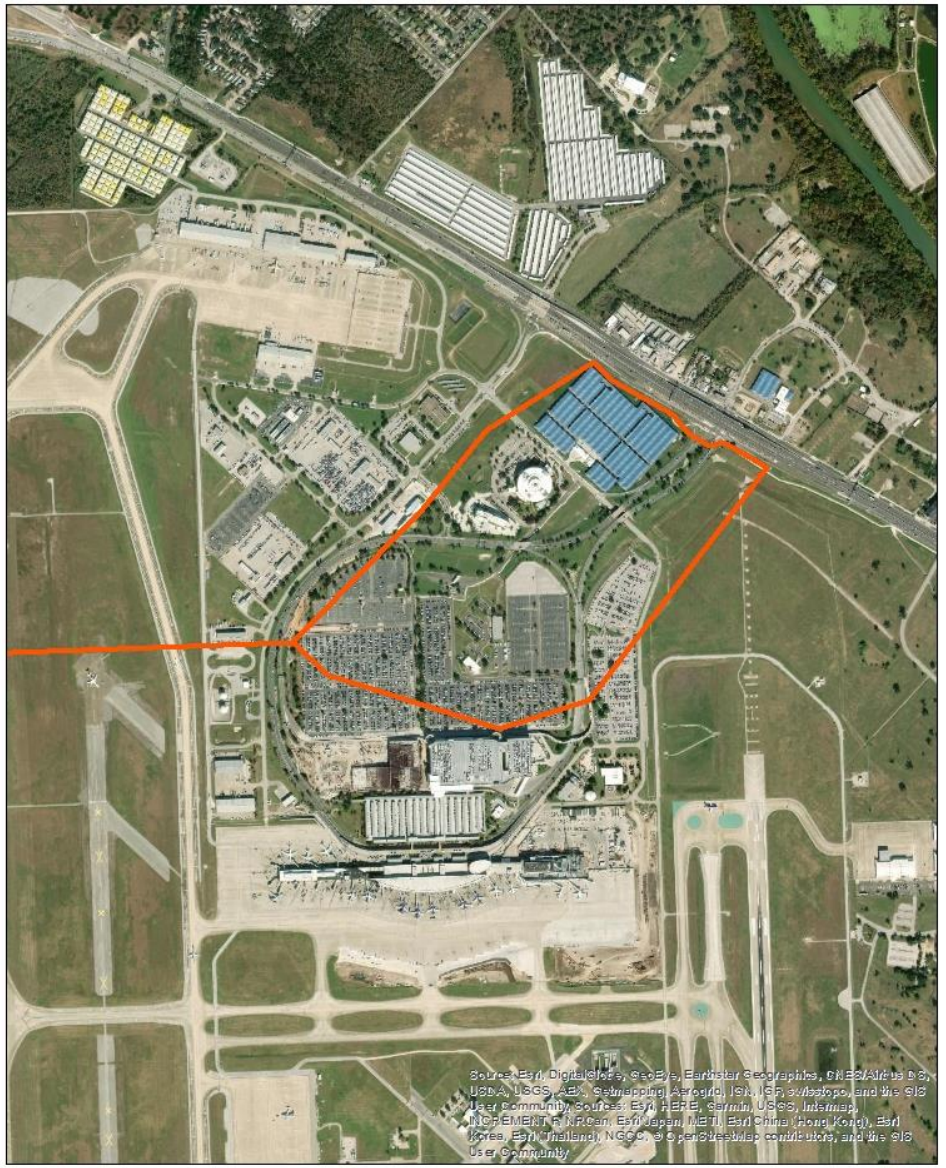
Data Entry

- Use coordinate of a known point to associate boundary
- Description called for end of runway and alignment of runway for “north”

BEARING BASIS NOTE

The bearings described herein are based on the Bergstrom Airport Coordinate System (BACS). The reference for this project is two brass disks in concrete found on the centerline of the west runway at the Austin – Bergstrom International Airport at an assumed bearing of NORTH.





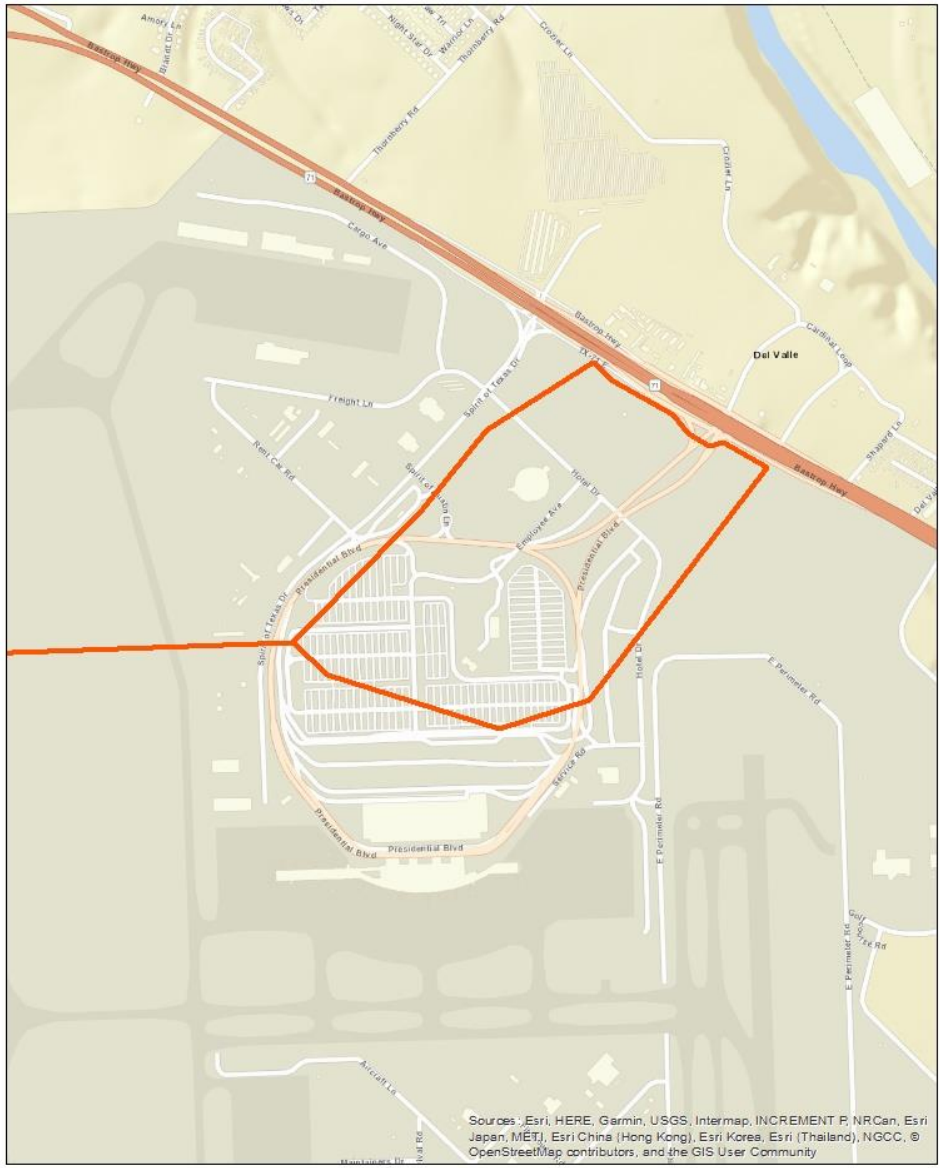
N

does not represent a boundary survey
boundary drafted with record description
Texas Coordinate System, South Central Zone, North American Datum, 1983

0 375 750 1,500 Feet

157.87 acres
City of Austin
dba Austin Airport Hilton

MEDINA VALLEY SURVEYS
Surveying Texas Since 1985
709 Washington Street, Castroville, TX 78009
MedinaValleySurveys.com 830.576.1634
TBPLS FIRM # 100002-00



does not represent a boundary survey
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Texas Coordinate System, South Central Zone, North American Datum, 1983

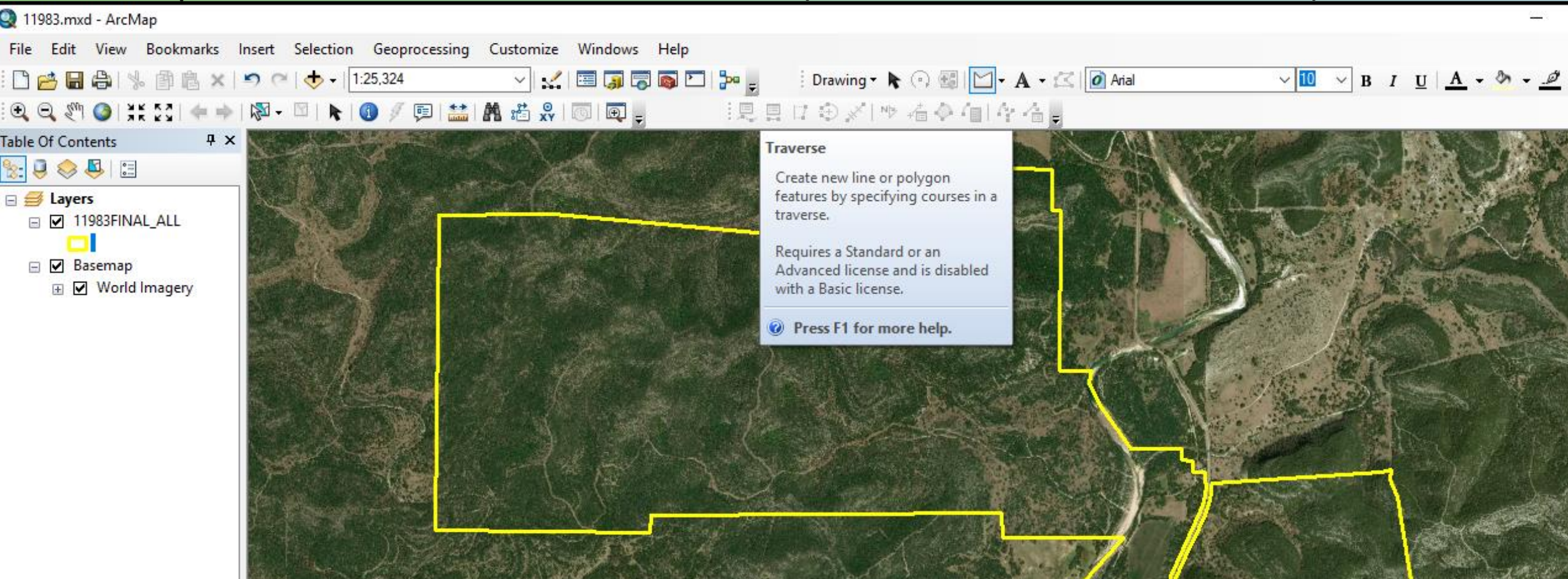
0 375 750 1,500 Feet

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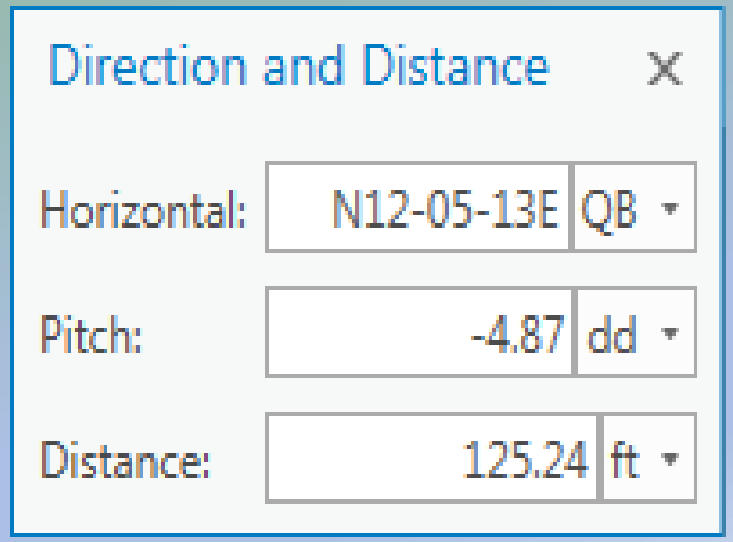
Draft a boundary:

- ArcGIS - Arcmap
- Toolbar – COGO - Traverse
- Requires Standard or Advanced license (Disabled with a Basic License)



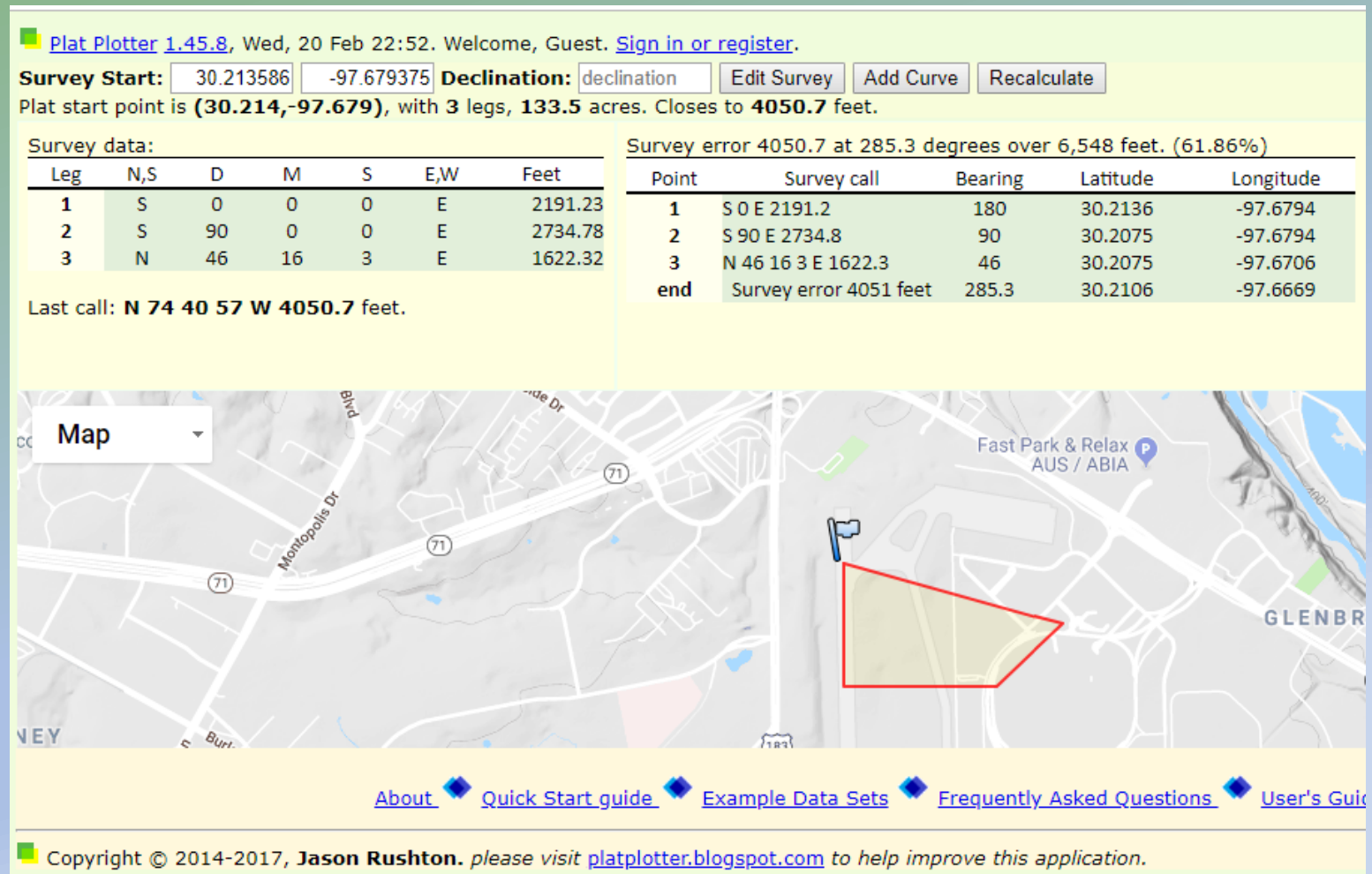
Draft a boundary:

- ArcGISPro
- After developing a feature class
- On the Edit tab, in the Features group, click Create
- The Create Features pane appears.
- Click a polyline or polygon feature template.
- Click the Line or Polygon tool.
- Create the first vertex using one of the following methods:
- Click the map.
- Right-click and click Absolute X,Y,Z , or press F6, type the values on the dialog box, and press Enter.
- Press G, or right-click and click Direction/Distance .
- The Direction and Distance dialog box appears.



Draft a boundary:

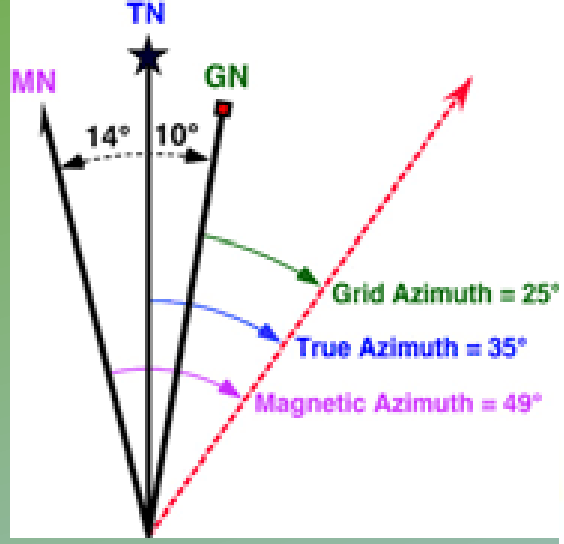
- Plat Plotter: http://platplotter.appspot.com
- Need start coordinate
- Enter with DMS
- Export boundary as KML

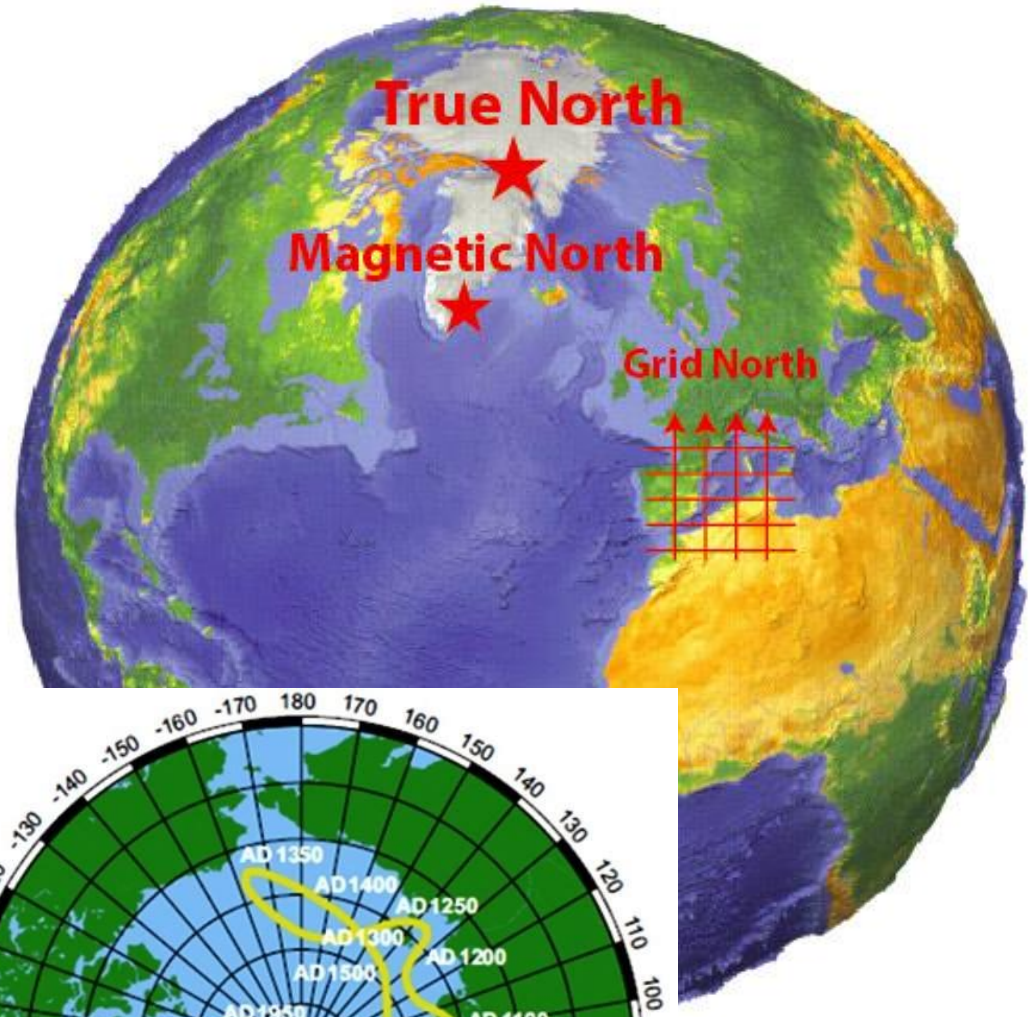


Things to consider

North

- astronomical north:
  is marked in the skies by the
  north celestial pole.



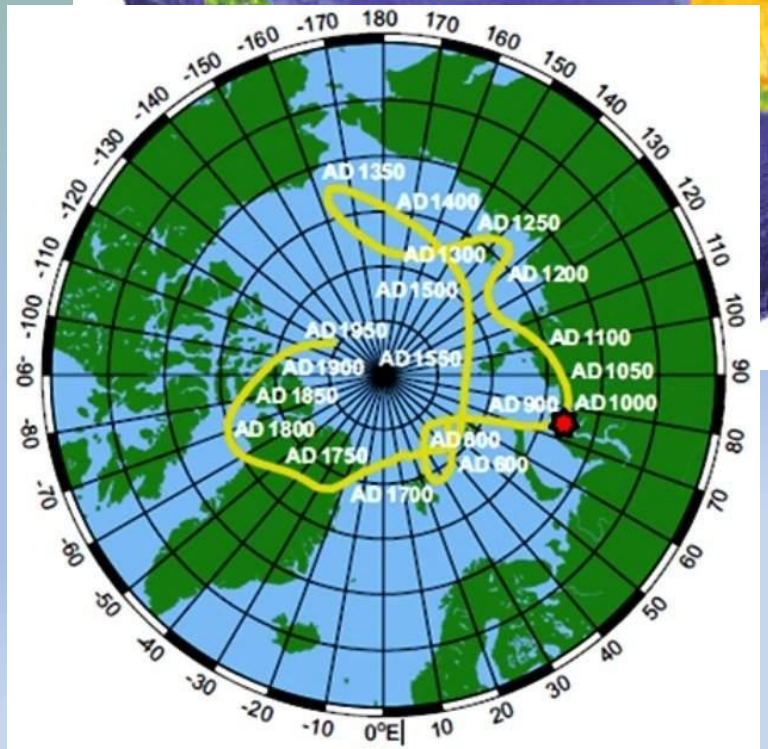


- geodetic north: a mathematical estimation of astronomic north derived by use of the “Laplace equation.”

- grid north: a navigational term referring to the direction northwards along the grid lines of a map projection.

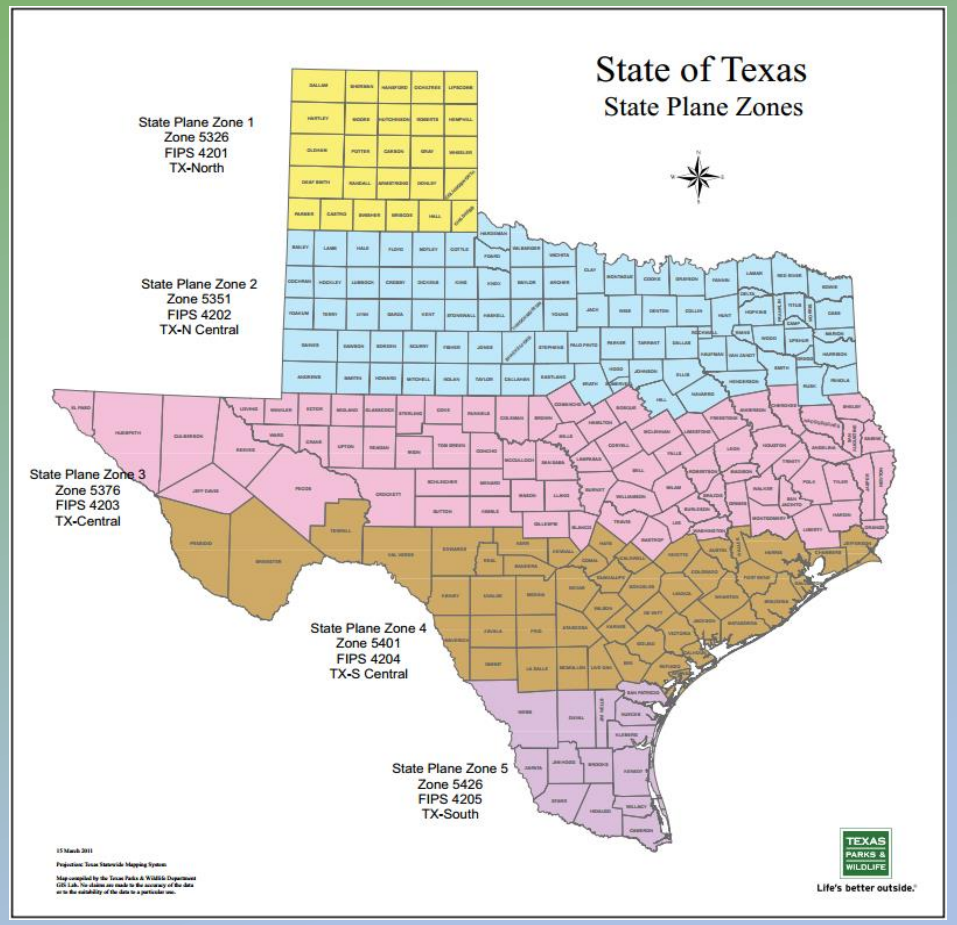
- magnetic north: the wandering point on the surface of Earth's Northern Hemisphere at which the planet's magnetic field points vertically downwards.

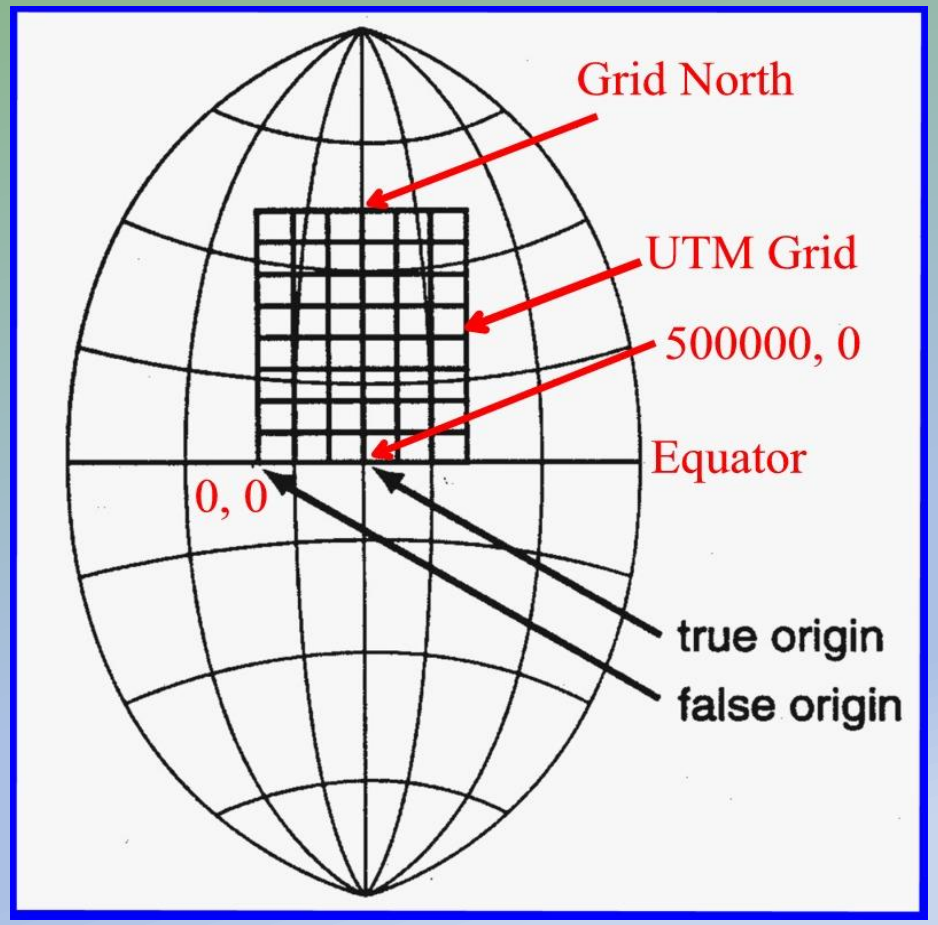
- true north: north that is an imaginary line through the Earth that leads to the North Pole.

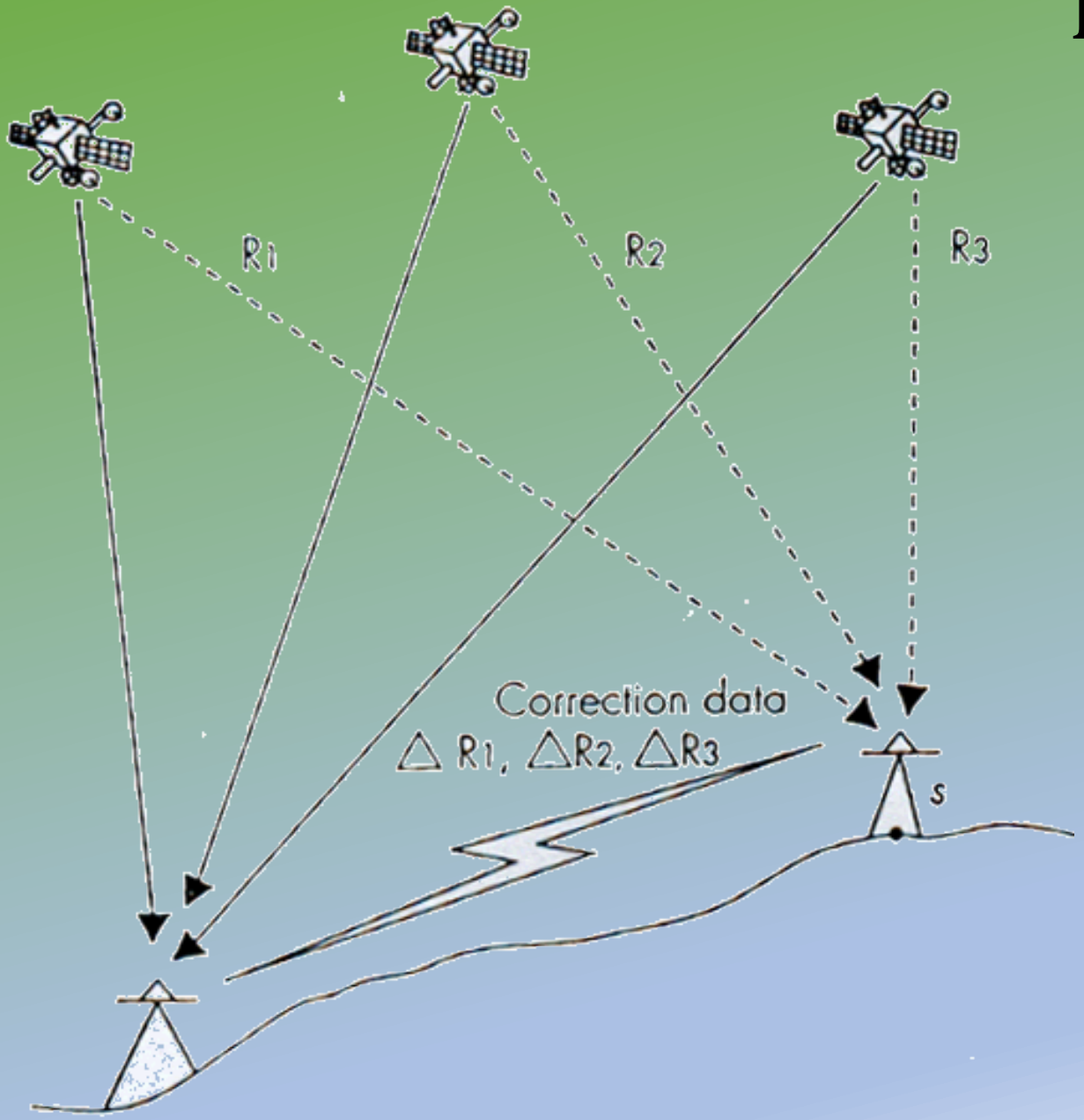


Coordinate Systems

- Projections state plan coordinates vs. geographic coordinate system







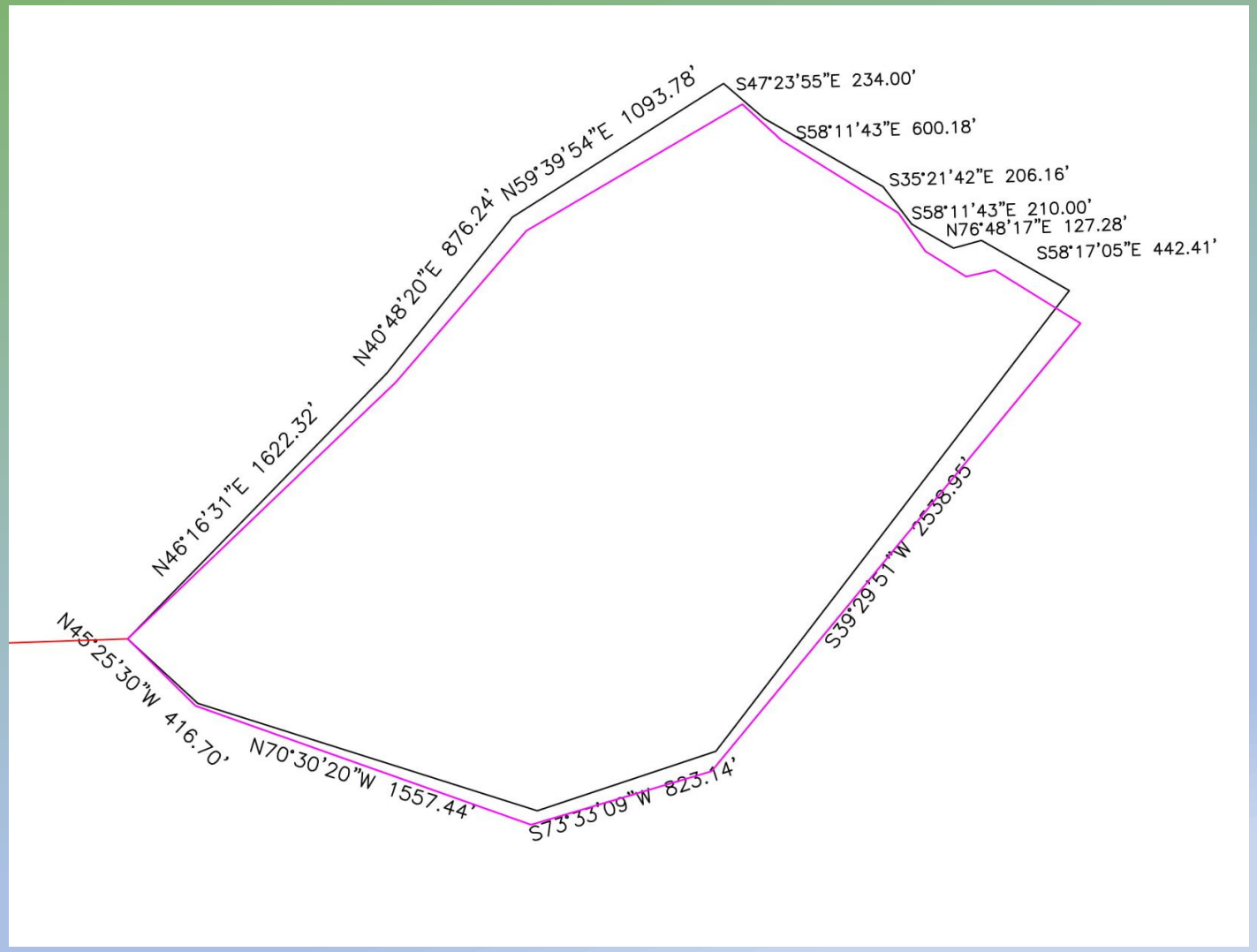
Data Collection:

GPS/GNSS/ RTK

- Differential GPS can achieve sub-centimeter accuracies in positions
- By placing a receiver at a known location, a total error factor can be computed and applied to the position data of the other receivers in the same locale.

Wrapping it up

- Orientation to “north” and coordinate system selection affects the bearing angle
- Rotation (without distortion) of boundary may be necessary to align with basemap/aerial image/ or field coordinates.





When to call for help

- A current survey with metes and bounds description should accompany every easement transaction.
- An RPLS needs to write all land descriptions (it's the law)
- Fences do not create boundary lines – real estate transaction do.